

L.J Institute of Engineering and Technology, Ahmedabad. Mathematics-I Practice Book_Batch 2025 & 2026_CE Related Branches								
Note : This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.								
Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
1	01	Simplify: $(x^{-3})^2 \div (x^{-2})^{-3}$.	x^{-12}	1	x^{-12}	x^{12}	x^0	x^6
2	01	Simplify: $\frac{x^{-7}}{x^{-8}} \cdot \frac{x^{-8}}{x^{-7}}$.	1	1	0	1	4	5
3	01	Simplify: $2x^2y \times 4x^5y$	$8x^7y^2$	1	$8x^7y^2$	$8x^2y^7$	$8x^3y^0$	$8x^{10}y^1$
4	01	Simplify: $\frac{2x^3y}{4xy^2}$	$\frac{x^2}{2y}$	1	$\frac{y^2}{2x}$	$\frac{x^2}{2y}$	$\frac{2x^2}{y}$	$\frac{x^3}{2y^2}$
5	01	Simplify: $\frac{x^4}{x^{-3}} \times \frac{x^1}{x^7}$	x	1	x	$\frac{1}{x}$	x^{25}	$\frac{1}{x^{25}}$
6	01	Find $(x^3)(x^{-5})(x^2)$	x^0	1	x^{10}	x^0	x^1	x^{-30}
7	01	Simplify: $\frac{7x^4}{14x^8}$	$\frac{1}{2x^4}$	1	$\frac{1}{2x^{32}}$	$\frac{1}{2x^4}$	$\frac{1}{2x^2}$	$\frac{1}{4}$
8	01	Simplify: $\left(\frac{2}{3}\right)^6 - \left(\frac{9}{4}\right)^{-3}$.	0	1	1	0	2	4
9	01	Find: $x^{-3} \times x^3$ at $x = 27$.	1	1	0	1	27	$\frac{1}{27}$
10	01	Simplify: $\frac{x^{-2018}}{x^{-2017}} \times \frac{x^3}{x^2}$	1	1	x^2	x^{-1}	x	1
11	01	Simplify: $\left(\frac{3^a}{3^b}\right)^{a+b} \cdot \left(\frac{3^b}{3^c}\right)^{b+c} \cdot \left(\frac{3^c}{3^a}\right)^{c+a}$		2				
12	01	Simplify: $\left(\frac{2}{3}\right)^6 \left(\frac{9}{4}\right)^{1/2} \div \left(\frac{27}{8}\right)^{-1/3}$		2				
13	01	Simplify: $2x^5y^4 \times 5x^3y^6$		2				
14	01	Simplify: $(2x^2y^{-3}) \times (\sqrt{16x^4y^6})$		2				
15	01	Find the value of $\left(x + \frac{1}{x}\right)^3$ at $x = \frac{1}{2}$.	$\frac{125}{8}$	1	$\frac{125}{8}$	$\frac{125}{16}$	$\frac{225}{8}$	$\frac{125}{18}$
16	01	Simplify: $\left(\frac{5}{7}\right)^2 \left(\frac{49}{25}\right)^{\frac{1}{2}} \div \left(\frac{5}{7}\right)^3 \left(\frac{125}{343}\right)^{\frac{-1}{3}}$		2				
17	01	Find value of $(x - x^{-1})^{-1}$ at $x = -1$	∞	1	0	5	∞	6
18	01	Find $\sqrt[4]{\sqrt{x}}$ at $x = 512$.		2				
19	01	Solve: $\sqrt{\sqrt{x}} = 2$.	16	1	9	16	10	12
20	01	If $x = \frac{a+y}{3}$ and $\frac{8^x}{2^y} = 64$ then find value of a .		2				
21	01	The value of $\sqrt[5]{x^4} \sqrt[4]{x^3} \sqrt[3]{x^2} \sqrt{x}$ is	$x^{\frac{119}{120}}$	1	$x^{\frac{119}{120}}$	$x^{\frac{121}{120}}$	$x^{\frac{119}{24}}$	$x^{\frac{17}{120}}$
22	01	If $3^{(x-y)} = 27$ and $3^{(x+y)} = 243$, then x is equal to	4	1	1	5	4	6
23	01	If $\left(\frac{3}{5}\right)^x = \frac{81}{625}$, then what is the value of x^x	256	1	16	32	0	256
24	01	If $(3^n)^3 \times 3^{3-n} = \frac{1}{27}$ then n=_____.	-3	1	3	2	9	-3
25	01	Simplify: $\sqrt{125} - 4\sqrt{6} + \sqrt{294} - 2\sqrt{\frac{1}{6}}$.		2				
26	01	If $x = \frac{2\sqrt{11} - 2\sqrt{7}}{\sqrt{11} + \sqrt{7}}$ and $y = \frac{3\sqrt{11} + 3\sqrt{7}}{\sqrt{11} - \sqrt{7}}$ then find $x^2 \cdot y^2$.		2				
27	01	If $\frac{2}{7+\sqrt{5}} = \frac{a}{b}(7 - \sqrt{5})$, then find a and b .		2				
28	01	If $a - b\sqrt{77} = \frac{\sqrt{11} - \sqrt{7}}{\sqrt{11} + \sqrt{7}}$ then find a and b .		2				

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29	01	If $x = 3 + 2\sqrt{2}$ then find the value of $(x - \frac{1}{x})^3$.		2				
30	01	Rationalize the denominator of $\frac{7+4\sqrt{3}}{7-4\sqrt{3}}$.		2				
31	01	Rationalize the denominator: $\frac{4+\sqrt{6}}{\sqrt{5}-\sqrt{4}}$		2				
32	01	Rationalize the denominator: $\frac{8+\sqrt{7}}{\sqrt{4}-\sqrt{3}}$		2				
33	01	Simplify: $\frac{\sqrt{3}}{\sqrt{75}+\sqrt{48}}$		2				
34	01	Rationalize the denominator of $\frac{4\sqrt{3}}{\sqrt{6} + \sqrt{2}}$.		2				
35	01	Simplify: $\frac{\sqrt{5}-1}{2+\sqrt{3}} + \frac{\sqrt{5}+1}{\sqrt{3}-2}$		2				
36	01	Simplify : $\frac{\sqrt{3}-1}{2+\sqrt{5}} + \frac{\sqrt{3}+1}{\sqrt{5}-2}$		2				
37	01	Simplify : $\frac{\sqrt{2}-1}{1+\sqrt{3}} + \frac{\sqrt{2}+1}{\sqrt{3}-1}$		2				
38	01	Rationalize the denominator $\frac{8+4\sqrt{3}}{8-4\sqrt{3}}$		2				
39	01	$\sqrt{5} + \sqrt{12 - 2\sqrt{35}}$ is equal to	$\sqrt{7}$	1	$\sqrt{7}$	$-\sqrt{7}$	$\sqrt{5}$	$\sqrt{7} + \sqrt{5}$
40	01	If $x = \frac{\sqrt{3}}{2}$ then find value of $\frac{\sqrt{1-x}}{1-\sqrt{1-x}} + \frac{\sqrt{1+x}}{1+\sqrt{1+x}}$.		3				
41	01	If $x = \frac{\sqrt{5}}{2}$ then $\frac{\sqrt{\frac{3+2x}{2}}}{2+\sqrt{\frac{3+2x}{2}}} + \frac{\sqrt{\frac{3-2x}{2}}}{2-\sqrt{\frac{3-2x}{2}}} =$ _____	$\frac{2\sqrt{5}}{5}$	1	$2\sqrt{5}$	$\frac{2\sqrt{5}}{5}$	1	$-\sqrt{5}$
42	01	Expand $(\log x^2 + x \log x)(7x - y \sin x^2)$.		2				
43	01	Expand: $(\log x - y^2 + \sin x)(x^2 + 2x)$.		2				
44	01	Factorize $4x^2 + 20x + 3yx + 15y$		2				
45	01	Factorise: $4x^2 + 15y + 3yx + 20x$.		2				
46	01	Factorize $x^2 - x - 6$	$(x - 3)(x + 2)$	1	$(x + 3)(x - 2)$	$(x - 3)(x - 2)$	$(x - 3)(x + 2)$	$(x + 3)(x + 2)$
47	01	Factorise: $6x^2 - 5x - 6$.		2				
48	01	Find values of x , if $5x^2 + 29x + 20 = 0$		2				
49	01	Find values of x , if $6x^2 + 5x - 6 = 0$		2				
50	01	Factories $3x^2 - 4x - 4$.		2				
51	01	Factorize $9x^2 - 12x - 5$.		2				
52	01	Solve for x : $x^2 - 3x - 10 = 0$		2				
53	01	Solve for x : $x^2 - 18x + 45 = 0$		2				
54	01	Solve for x : $x^2 + 5x + 6 = 0$		2				

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55	01	Solve for x : $x^2 - 3x - 4 = 0$		2				
56	01	Factorize: $x^2 - 23x + 120$		2				
57	01	Factorize: $x^2 + 9x - 22$		2				
58	01	Factorise: $x^2 - 5x - 6$		2				
59	01	Factorise: $x^2 + 5x - 6$		2				
60	01	Factorise: $2x^2 + 2x + 3x + 3$		2				
61	01	Factorize: $2x^2 - 128$.		2				
62	01	What is the value of p if $x + 1/2$ is a factor of $px^2 - 4px + 3$?	-4/3	1	-4/3	7	2	-1/2
63	01	Factorize: $54x^3 - 2$.		2				
64	01	Solve: $(x^3 - 1) = 0$.		2				
65	01	Factorise: $x^3 + 3x^2 - 4x - 12$.		2				
66	01	Factorize $x^3 - 18x^2 + 99x - 162$		2				
67	01	Find the all real roots of the equation $4x^3 - 108 = 0$.		2				
68	01	Factorize the polynomial $2x^3 + 9x^2 + 10x + 3$.		3				
69	01	Factorize the polynomial $x^3 - 6x^2 - x + 30$.		3				
70	01	Factorize the polynomial $x^3 + x^2 - 21x - 45$.		3				
71	01	Factorize: $x^3 - 4x^2 - x + 4$		3				
72	01	Factorize: $x^3 - 6x^2 + 11x - 6$		3				
73	01	Factorize $x^3 - 4x^2 + x + 6 = 0$		3				
74	01	Factorize: $x^3 - 4x^2 - 7x + 10$		3				
75	01	Factorize: $36x^3 + 30x^2 - 36x$.		3				
76	01	Factorize : $-x^3 + 8x^2 - 19x + 12$		3				
77	01	Factorize: $x^3 - 5x^2 + 8x - 4$		2				
78	01	Factorize: $270x^3 - 10$		2				
79	01	Which of the following is equal to $x^3 - 225x$?	$x(x - 15)(x + 15)$	1	$x(1 - 15x)(1 + 15x)$	$x(1 - 15x)(1 - 15x)$	$x(1 + 15x)(1 + 15x)$	$x(x - 15)(x + 15)$
80	01	The roots of the equation $x^3 - 8x^2 + 17x - 10 = 0$ are		2				
81	01	Solve: $x^4 + x^3 - 3x^2 - x + 2 = 0$		2				
82	01	Solve: $3x^4 - 8x^3 - 37x^2 + 2x + 40 = 0$		2				
83	01	Factorize $x^4 - 81y^4$		2				
84	01	Factorize $x^4 - 256y^4$		2				

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85	01	Find the all real roots of the equation $x^4 - 256 = 0$.		2				
86	01	Factorize the polynomial $2x^4 + 7x^3 - 29x^2 - 112x - 48$		3				
87	01	Factorize: $x^4 - 7x^3 - 5x^2 + 111x - 180$		2				
88	01	Find n_{C_n} .	1	1	0	1	2	4
89	01	Find n_{C_0} .	1	1	0	1	2	4
90	01	Find $\binom{10}{8}$	45	1	0	1	45	4
91	01	Find $\binom{25}{5}$	53130	1	53130	53132	0	52345
92	01	Find $\binom{12}{0} + \binom{12}{1} + \binom{12}{2} + \dots + \binom{12}{12}$		2				
93	01	Write down the number of terms in the expansion of $(x - \frac{1}{x})^n$.	n+1	1	2n+1	n+1	7n+1	6n+1
94	01	Write formulas for following . (a) $(a + b)^3$ (b) $a^3 + b^3$ (c) $(a - b)^2$ (d) $a^3 - b^3$		2				
95	01	If $a^2 + b^2 = 5$ and $ab = \frac{3}{2}$, then find $a - b = ?$		2				
96	01	Find $15C_4$		2				
97	01	Write formulas for following . (a) $(a - b)^3$ (b) $a^3 - b^3$ (c) $(a + b + c)^2$ (d) n_{C_r}		2				
98	01	Expand $(1 - y)^5$ using binomial theorem.		2				
99	01	Expand $(x - y)^3$ using binomial theorem .		2				
100	01	Find the value of $(x - \frac{1}{x})^3$ at $x = 2$.	$\frac{27}{8}$	1	$\frac{26}{8}$	$\frac{27}{8}$	$\frac{25}{8}$	$\frac{23}{8}$
101	01	Expand $(a - b - c)^2$		2				
102	01	Expand $(a + b)^2$ using binomial theorem .		2				
103	01	Expand $[5x - \frac{1}{5y}]^5$ using Binomial Theorem.		2				
104	01	If $(3 + 2\sqrt{5})^2 = 29 + K\sqrt{5}$, then the value of $K = \underline{\hspace{1cm}}$	12	1	12	6	29	39
105	01	The fourth term in the expansion $(x - 2y)^{12}$ is _____	$-1760 x^9 y^3$	1	$-1670 x^9 y^3$	$-7160 x^9 y^3$	$-1760 x^9 y^3$	$-1607 x^9 y^3$
106	01	The coefficient of x^6 in the series expansion of $(x + 2)^9 = \underline{\hspace{1cm}}$	672	1	672	762	627	267
107	01	Expand $\left(\frac{3x^2}{2} - \frac{1}{3x}\right)^7$ using binomial theorem.		2				
108	01	Expand: $(-7+5x)^6$		2				
109	01	Using binomial theorem expand $(3x + 2y)^5$.		2				
110	01	Using binomial theorem expand $(x - 2)^6$ and find coefficient of x^4 .		2				

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111	01	Using binomial theorem expand $(2x + 3y)^5$		2				
112	01	Using binomial theorem expand $(x - y)^6$ and find the coefficient of x^3y^3		2				
113	01	Write the formula of $(1 - b)^n$.		2				
114	01	What is the coefficient of x in expansion of $\left(3x + \frac{1}{x}\right)^3$?		2				
115	01	What is the constant term in expansion of $\left(x - \frac{1}{x}\right)^4$?		2				
116	01	Using Binomial Theorem expand: $(2x + 3)^4$		2				
117	01	Using Binomial Theorem expand: $(x - 1)^5$		2				
118	01	Which is a mid-term of in the expansion of $\left(\frac{2x}{9} - \frac{3}{2}\right)^8$?	$\frac{70x^4}{81}$	1	$\frac{1}{81}$	$\frac{56}{81}$	$\frac{70x^4}{81}$	$\frac{x^4}{81}$
119	01	Find the middle term in expansion of $\left(2x + \frac{1}{x}\right)^5$	80x and 40/x	1	80x and 40/x	40x and 40/x	80x and 80/x	80 and 40/x
120	01	Expand $(a - 2b)^4$ using binomial expansion.		2				
121	01	Using binomial theorem expand $(x - y)^8$		2				
122	01	Using Binomial Theorem expand: $\left(\sqrt{2x} + \frac{1}{\sqrt{2x}}\right)^4$.		2				
123	01	What is the coefficient of x^3 , from the expansion of $(x^2 - 3)^4$.		2				
124	01	Expand: $(2x - 1)^5$		2				
125	01	Use binomial theorem to expand $(a + b)^4$.		2				
126	01	Use binomial theorem to find $a^3 - b^3$.		2				
127	01	Using Binomial Theorem expand: $(x - 2)^5$		2				
128	01	Using Binomial Theorem expand: $(3x + 2)^5$		2				
129	01	Using Binomial Theorem expand: $(3x - 2)^5$.		2				
130	01	The coefficient of x^2y^3 in expansion of $(2x - 3y)^5$ is ____	-1080	2	-1080	$\frac{27}{8}$	-343	27
131	01	Expand $\left[3x + \frac{1}{3x}\right]^4$ using Binomial Theorem .		2				
132	01	Find Middle Term of $\left(\frac{x}{2} + 3y\right)^8$		3				
133	01	Expand: $(-7 + 5x)^6$		2				
134	01	Expand: $(-7 + 7x)^6$		2				
135	01	Find the coefficient of x^5 in the expansion of $(3x - 5)^{10}$		2				
136	01	Simplify $\log_a 4 + \log_a 2$	$\log_a 8$	1	$\log_a 16$	$\log_a 2$	$\log_a 8$	$2\log_a 3$

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137	01	If $\log_5 x = 3$,then find x .	125	1	125	243	15	$\frac{1}{125}$
138	01	Find $\log_{10}10$	1	1	0	1	10	e
139	01	If $\log_2(a) = 3$ then find a .	8	1	8	9	6	1.5
140	01	Find value of $\log 0$.	$-\infty$	1	$-\infty$	∞	0	1
141	01	If $\log_3 x = -7$,then find x .	$\frac{1}{2187}$	1	-343	343	$\frac{1}{2187}$	$-\frac{1}{2187}$
142	01	If $2 \ln x = 1$ then $x =$ _____	$\sqrt{e}$	1	e	e^2	$\sqrt{e}$	10
143	01	If $\log_x \left(\frac{9}{16}\right) = -\frac{1}{2}$, then $x =$ _____	$\frac{256}{81}$	1	$\frac{256}{81}$	$-\frac{3}{4}$	$\frac{3}{4}$	$\frac{81}{256}$
144	01	Find value of $\log \infty$.	∞	1	$-\infty$	∞	0	1
145	01	Simplify $\log_a 4 - \log_a 2$	$\log_a 2$	1	$\log_a 16$	$\log_a 2$	$\log_a 8$	$2\log_a 3$
146	01	If $\log_{10} 7 = a$, then $\log_{10} \left(\frac{1}{70}\right)$ is equal to	$-(1 + a)$	1	$(1 + a)^{-1}$	$\frac{a}{10}$	$-(1 + a)$	$\frac{1}{10a}$
147	01	If $\log_2 (x^2 - 6x) = 3 + \log_2 (1 - x)$ then value of x is	-4	1	-4	-3	4	-2
148	01	If $\log_x \left(\frac{9}{16}\right) = -\frac{1}{2}$, then $x =$ _____	$\frac{256}{81}$	1	$\frac{256}{81}$	$-\frac{3}{4}$	$\frac{3}{4}$	$\frac{81}{256}$
149	01	$\frac{\log x + \log x^4 + \log x^9 + \log x^{16} + \dots \log x^{169}}{\log x + \log x^2 + \log x^3 + \log x^4 + \dots \log x^{13}}$ =	9	1	9	0	3	5
150	01	Simplify: $\log \frac{x^2}{y^2} + \log \frac{y^2}{z^2} - \log \frac{x^2}{z^2}$.		2				
151	01	Simplify: $\log x + \log x^2 + \log x^3 + \log x^4 + \dots + \log x^{100}$ at $x = 1$.		2				
152	01	If $\log \frac{1}{x} - \log \frac{1}{y} = 0$ then find $(x - y)^3$.		2				
153	01	If $\log 5 = 0.6990$ and $\log 3 = 0.4771$ then find $\log 15$.		2				
154	01	Find $\log_2 1024$		2				
155	01	Find $\log_2 128$		2				
156	01	Find $\log x + \log x^2 + \log x^3 + \log x^4$ at $x = e$.		2				
157	01	$\log_6(x + 4) + \log_6(x - 2) = \log_6 4x$ Find x .		2				
158	01	If $\log_4 x = 3$ Find x .		2				
159	01	Simplify : $\log_8 2$		2				
160	01	Solve for x : $2(\log x)^2 + 5 \log x = 3$		2				
161	01	If $\log_2 3. \log_3 4. \dots \log_n (n + 1) = 10$ then find value of n.		2				
162	01	Prove that $\frac{1}{\log_6 24} + \frac{1}{\log_{12} 24} + \frac{1}{\log_8 24} = 2$		2				

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163	01	If $\log_3 x = 2$ Find x		2				
164	01	Find $\log x . \log 2x . \log 3x \dots \log 10x$ at $x = 1$.		2				
165	01	Solve : $\log_3(9x + 2) = 4$.		2				
166	01	Solve for x : $\log_2(x + 1) - \log_2(x - 4) = 3$		2				
167	01	$\log x + \log x^{2^2} + \log x^{3^2} + \dots + \log x^{22^2} =$ _____, for $x = e$.	3795	2	253	506	3795	$\frac{3795}{2}$
168	01	Solve: $\log_2 x - \log_4 3 = 2$		2				
169	02	Find $\operatorname{cosec}\left(\frac{5\pi}{2} + \theta\right)$	$\sec \theta$	1	$\operatorname{cosec} \theta$	$\sec \theta$	$-\sec \theta$	$-\operatorname{cosec}$
170	02	Find $\cot(5\pi - \theta)$	$-\cot \theta$	1	$-\tan \theta$	$\tan \theta$	$-\cot \theta$	$\cot \theta$
171	02	Find the value of $\cos 135^\circ$.		2				
172	02	Find: $\frac{\sin 72^\circ}{\cos 18^\circ} + \frac{\cos 72^\circ}{\sin 18^\circ} - 2 \sin 90^\circ$.		2				
173	02	Find $\operatorname{Sec}(8\pi + \theta)$	$\sec \theta$	1				
174	02	Find $\tan\left(\frac{9\pi}{2} - \theta\right)$	$\cot \theta$	1				
175	02	Find the value of $\sin(75^\circ)$		2				
176	02	Find $\cos(3\pi - \theta)$	$-\cos \theta$	1				
177	02	Find $20 \sin 330^\circ - 15 \sin 30^\circ$		2				
178	02	Identify even function from following functions: [a] $\sin(x)$ [b] $\cos(x)$ [c] $\sin x \cos x$ [d] $\sin x + \cos x$	$\cos x$	1	$\sin x$	$\cos x$	$\sin x \cos x$	$\sin x + \cos x$
179	02	$\sin 2A = 2 \sin A$ is true then $A = ?$	0°	1	0°	30°	45°	60°
180	02	If $4 \tan A = 3$, then $\frac{4 \sin A - \cos A}{4 \sin A + \cos A} = ?$	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{1}{3}$	$\frac{3}{4}$
181	02	If $\sin A = \frac{1}{2}$ and $\cos B = \frac{1}{2}$, then $A + B = ?$	90°	1	0°	30°	60°	90°
182	02	If $\sec A + \tan = x$, then $\sec A = ?$	$\frac{x^2 + 1}{2x}$	1	$\frac{x^2 - 1}{x}$	$\frac{x^2 + 1}{x}$	$\frac{x^2 - 1}{2x}$	$\frac{x^2 + 1}{2x}$
183	02	If $\sec A + \tan A = x$, then $\tan A = ?$	$\frac{x^2 - 1}{2x}$	1	$\frac{x^2 + 1}{2x}$	$\frac{x^2 - 1}{2x}$	$\frac{x^2 - 1}{x}$	$\frac{x^2 + 1}{x}$
184	02	If $\operatorname{cosec} A - \cot A = \frac{1}{3}$, then $\operatorname{cosec} A + \cot A = ?$	3	1	1	2	3	4
185	02	If $\frac{2 \sin x - \cos x}{\cos x + \sin x} = 1$ then value of $\cot x$ is _____.	$\frac{1}{2}$	1	$-\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{2}$	$-\frac{1}{3}$
186	02	Write down the maximum and minimum value of $5 \sin \theta + 3$.	Max Value:8 Min Value:-2	1				
187	02	If $\operatorname{cosec} \theta = \sec \theta$ then find $\tan \theta$.	1	1	1	2	4	5
188	02	Find Maximum value of $5 \cos(x) + 12 \sin(x) + 12$		2				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
189	02	If $\tan \theta = 1$ then Find $\operatorname{cosec} \theta$.($0 < \theta < \pi/2$)	$\sqrt{2}$	1	$\sqrt{2}$	$2\sqrt{2}$	5	6
190	02	Check whether $x^2 \sin^2 x$ is an even or odd function.		2				
191	02	Prove that $\tan \theta$ is an odd function.		2				
192	02	Prove that $\frac{\tan \theta}{1-\cot \theta} + \frac{\cot \theta}{1-\tan \theta} = 1 + \tan \theta + \cot \theta$		2				
193	02	The value of $(\tan 1^\circ \tan 2^\circ \tan 3^\circ \dots \tan 89^\circ)$	1	1	0	1	2	$\frac{1}{2}$
194	02	The value of $\frac{2 \tan 30^\circ}{1+\tan^2 30^\circ}$	$\sin 60^\circ$	1	$\sin 60^\circ$	$\cos 60^\circ$	$\tan 60^\circ$	$\sin 30^\circ$
195	02	The value of $\frac{2 \tan 30^\circ}{1-\tan^2 30^\circ}$	$\tan 60^\circ$	1	$\sin 60^\circ$	$\cos 60^\circ$	$\tan 60^\circ$	$\sin 30^\circ$
196	02	If $\cos(A+B)=0$, then $\sin(A-B)=?$	$\cos 2B$	1	$\cos A$	$\cos 2B$	$\sin A$	$\sin 2$
197	02	If $\tan 3x = \sin 45^0 \cos 45^0 + \sin 30^0$ then $x = \underline{\hspace{1cm}}$.	15^0	1	30^0	10^0	15^0	45^0
198	02	If $\sin(A+B)=0$, then find $\cos(A-B)$.		2				
199	02	$\sin(45^\circ + \theta) - \cos(45^\circ - \theta) = ?$	0	1	$2 \cos \theta$	0	$2 \sin \theta$	1
200	02	If $\tan(A+B)=3$ and $\tan(A-B)=2$ then find $\tan 2A$ and $\tan 2B$.		3				
201	02	Derive the formula of $\sin(A-B)$ with the help of $\sin(A+B)$.		2				
202	02	Find: $2 \sin \frac{5\pi}{12} \cdot \cos \frac{\pi}{12}$		2				
203	02	Derive the formula of $\cos 2x$ in terms of $\cos x$.		2				
204	02	Derive the formula of $\tan 3x$ in terms of $\tan x$.		2				
205	02	Prove that $\cos 3x = 4 \cos^3 x - 3 \cos x$		2				
206	02	Prove that $\tan(2x) = \frac{2 \tan x}{1-\tan^2 x}$		2				
207	02	Prove that $\sin 3x = 3 \sin x - 4 \sin^3 x$		2				
208	02	Prove: $\cos(2x) = \cos^2 x - \sin^2 x$		2				
209	02	Prove: $\cos(2x) = 2 \cos^2 x - 1$		2				
210	02	Find: $\sin \frac{\pi}{16} \cos \frac{\pi}{8} \cos \frac{\pi}{4} \cos \frac{\pi}{16}$		2				
211	02	Prove that $\frac{\sin A}{1+\cos A} = \tan \frac{A}{2}$		2				
212	02	Find value of $\sin 75^\circ \cos 15^\circ + \sin 15^\circ \cos 75^\circ$		2				
213	02	Find value of $\sin 75^\circ \sin 15^\circ$.		2				
214	02	Prove that $\cos 2x = \frac{1-\tan^2 x}{1+\tan^2 x}$		2				
215	02	Find: $\sin 89^\circ \cdot \operatorname{cosec} 89^\circ + \tan 89^\circ \cdot \cot 89^\circ + \cos 89^\circ \cdot \sec 89^\circ$		2				
216	02	Derive the formula of $\sin 2x$.		2				
217	02	Find: $\sin 70^\circ \cdot \cos 20^\circ + \cos 70^\circ \cdot \sin 20^\circ$.		2				
218	02	Find value of $1 - 2 \sin^2 90^\circ$		2				
219	02	Find value of $\tan 15^\circ$		2				
220	02	Find value of: $8 \cos 15^\circ \cos 45^\circ \cos 75^\circ$		2				
221	02	Derive Formula of $\tan(\alpha + \beta)$		2				
222	02	Derive Formula of $\cot(\alpha + \beta)$		2				
223	02	a) Derive Formula of $\tan(\alpha - \beta)$ b) Derive the formula of $\tan(A+B)$ with the help of $\tan(A-B)$.		2				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
224	02	Find value of $\cos 70^\circ \cos 20^\circ - \sin 70^\circ \sin 20^\circ$		2				
225	02	Find value of: $8 \sin 15^\circ \sin 45^\circ \sin 75^\circ$		2				
226	02	Find $16 \sin^3 40^\circ - 12 \sin 40^\circ$.		2				
227	02	Write the formula of $\tan^{-1}x - \tan^{-1}y$.		2				
228	02	In ΔABC $B = 21^\circ$, $C = 46^\circ$ and $AB = 9$ cm.Solve this triangle.		2				
229	02	In ΔABC $A = 42^\circ$, $BC = 37$ cm and $AC = 26$ cm.Solve this triangle.		2				
230	03	Find $\frac{d}{dx}(x^3)$	$3x^2$	1	$3x^2$	$\frac{x^4}{4}$	$2x^3$	$\frac{4}{x^4}$
231	03	What is the value of $\frac{dy}{dx}$ if $x - y = 1$?	1	1	1	-1	0	2
232	03	$\frac{d}{dx}(\sec^{-1}(3x)) =$ _____	$\frac{1}{ x \sqrt{9x^2 - 1}}$	1	$\frac{1}{ x \sqrt{9x^2 - 1}}$	0	$-\frac{1}{ x \sqrt{9x^2 - 1}}$	$-\frac{3}{ x \sqrt{9x^2 - 1}}$
233	03	If $f(x) = \sqrt{x}$ then $\frac{f''(x)}{f'(x)} =$ _____.	$-\frac{1}{2x}$	1	$\frac{1}{2x}$	$2x$	$-2x$	$-\frac{1}{2x}$
234	03	Find $\frac{d}{dx}(\tan^{-1}x + \tan^{-1}y)$. If $xy = 1$.	0	1	0	1	2	3
235	03	Find $\frac{d}{dx}(11 \cot^2 x - 11 \operatorname{cosec}^2 x)$.		2				
236	03	Find $\frac{d}{dx}(2^x)$		2				
237	03	Find $\frac{d}{dx}(\log_{10} x)$		2				
238	03	Find $\frac{d}{dx}(5a^x + 3x^a - 5e^{\frac{x}{2}})$		2				
239	03	Find $\frac{d}{dx}(\sin^2 x + \cos^2 x)$.		2				
240	03	Find $\frac{d}{dx}(\operatorname{cosec} x)$	$-\operatorname{cosec} x \cot x$	1				
241	03	Find $\frac{d}{dx}(a^x)$	$a^x \log_e a$	1				
242	03	Find $\frac{d}{dx}(\cot x)$	$-\operatorname{cosec}^2 x$	1				
243	03	Find $\frac{d}{dx}(e^x)$	e^x	1				
244	03	If $f(x) = ax^2 + bx + 12$, $f'(4) = 15$ and $f'(2) = 11$ then find a and b .		3				
245	03	Find $\frac{d}{dx}(e^{3x} \tan 2x)$		2				
246	03	Find $\frac{d}{dx}\left(\frac{\log x}{\sin x}\right)$		2				
247	03	Find $\frac{d}{dx}(\operatorname{cosec} x \log_e \sec x)$		2				
248	03	Find $\frac{d}{dx}\left(\frac{e^x - e^{-x}}{e^x + e^{-x}}\right)$		2				
249	03	Find $\frac{d}{dx}\left(\frac{\cos x^3}{\cos^3 x}\right)$		2				
250	03	Find $\frac{d}{dx}(2^x \operatorname{cosec}^3 x)$		2				

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251	03	Find the derivative of $x^2 \tan^{-1} x$.		2				
252	03	Find $\frac{d}{dx} \left(\frac{\log 5x}{\tan x} \right)$		2				
253	03	Find $\frac{d}{dx} (\sin^4 3x \tan^3 4x)$		2				
254	03	Find $\frac{d}{dx} \left(\frac{\sec x}{x^2 + 1} \right)$		2				
255	03	Find $\frac{d}{dx} (\sin x^2 \log 3x)$		2				
256	03	Find derivative of $\sin(\cos x^5) x^5$ with respect to x		2				
257	03	Find derivative of $\frac{\sin(e^{x^2})}{e^{\sin(x^2)}}$ with respect to x		2				
258	03	Find $\frac{d}{dx} \left(\frac{\cot^4 x}{5x^2 - 4} \right)$		2				
259	03	Find $\frac{d}{dx} (\log 6x \operatorname{cosec}^4 4x)$		2				
260	03	Write down the product and division rule for derivative of f(x) and g(x).		2				
261	03	Find: $\frac{d}{dx} (\sin^3 x \cdot \sin x^3)$.		2				
262	03	Find $\frac{d}{dx} \left\{ (x + 1)(\sqrt{x} - 1) \left(\frac{1}{x} + 4 \right) \right\}$		2				
263	03	Find $\frac{d}{dx} \left(\frac{\operatorname{cosec}^4 x}{25x^2 - 4} \right)$		2				
264	03	Find: $\frac{d}{dx} (x \sin x)$		2				
265	03	Find: $\frac{d}{dx} \left(\frac{x}{\tan x} \right)$		2				
266	03	Find $\frac{d}{dx} (e^{3x} \sin 3x \log 3x)$.		2				
267	03	Find $\frac{d}{dx} (\sin (\cos 3x) \cdot e^{3x})$.		2				
268	03	Find $\frac{d}{dx} (\sec(5x) \log_e x^4)$		2				
269	03	Find $\frac{d}{dx} \left(\frac{a^x}{\sin x} \right)$		2				
270	03	Find: $\frac{d}{dx} (\cos^3 x \cdot \cos x^3)$.		3				
271	03	Find $\frac{d}{dx} (\sin^3 x)$	$3\sin^2 x \cos x$	1	$3\cos^2 x \sin x$	$\frac{\sin^4 x}{4}$	$3\sin^2 x \cos x$	$\frac{\cos^4 x}{4}$
272	03	$\frac{d}{dx} (\sin x^x) =$ _____	$(\cos x^x) x^x (1 + \log x)$	1	$(\cos x^x) x^x (1 + \log x)$	$-(\cos x^x) (1 + \log x)$	$-(\cos x^x) (1 + \log x)$	$(\cos x^x) (1 + \log x)$
273	03	Find derivative of $f(x) = \sqrt[3]{\sin^3 x}$ with respect to x		2				
274	03	Find $\frac{d}{dx} \sqrt{\sin x^3}$		2				
275	03	Find $\frac{d}{dx} \tan(2x^2 + 2x + 2)$		2				
276	03	Find $\frac{d}{dx} \sec^2 7x$		2				
277	03	Find: $\frac{d}{dx} (e^{\log \sqrt{x}})$		2				
278	03	Find: $\frac{d}{dx} (\sin^{-1} x + \cos^{-1} x)$		2				
279	03	Find $\frac{d}{dx} (e^{\sqrt{3x-9}})$		2				

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280	03	Find $\frac{d}{dx}(a^{\sin x})$		2				
281	03	Find $\frac{d}{dx}(\tan x^3)$		2				
282	03	Find $\frac{d}{dx}(\log \sec x)$		2				
283	03	Find $\frac{d}{dx}(\log_{10} x^3 + \sin x^o)$		2				
284	03	Find $\frac{d}{dx}(\sec^{-1} 2x)$		2				
285	03	Find $\frac{d}{dx}(e^{3x+1})$		2				
286	03	Find $\frac{d}{dx}(\sin^{-1} 3x)$		2				
287	03	Find $\frac{d}{dx}(\log \operatorname{cosec} x^o)$		2				
288	03	Find $\frac{d}{dx}(\sin^4 3x \tan^3 4x)$.		2				
289	03	Find $\frac{d}{dt}(t^t)$.		2				
290	03	Find $\frac{d}{dx} \frac{(\sin x^3 \tan \sqrt{x})}{\operatorname{cosec} x}$.		2				
291	03	Find: $\frac{d}{dx}(\sec x^x)$		2				
292	03	Find: $\frac{d}{dx}(\cos x^x)$		2				
293	03	Find: $\frac{d}{dx}(2019 \cot x^x)$		2				
294	03	Find: $\frac{d}{dx}(\tan x^x)$		2				
295	03	It is given that Roll's Theorem holds good for the function $f(x) = x^3 + ax^2 + bx, x \in [1,2]$ at the point $x = \frac{4}{3}$. Find Values of a and b.		3				
296	03	Verify Roll's Theorem for $f(x) = e^{-x} \sin x$ on $[0, \pi]$		3				
297	03	Verify Roll's Theorem for $f(x) = e^{2x}(\sin 2x - \cos 2x)$ on $\left[\frac{\pi}{8}, \frac{5\pi}{8}\right]$		3				
298	03	Verify Lagrange's mean value theorem for the function $f(x) = \sqrt{x^2 - x}$ in the interval $[1,4]$.		3				
299	03	Find 'c' using Lagrange's Mean Value theorem if $f(x) = x^2 - 3x - 1, x \in \left[-\frac{11}{7}, \frac{13}{7}\right]$.		3				
300	03	Find Value of c using Cauchy's mean value theorem in $[1,2]$ if $f(x) = e^x$ and $g(x) = e^{-x}$.		3				
301	03	Verify Cauchy's mean value theorem for $f(x) = \sin x$ and $g(x) = \cos x$ in $\left[0, \frac{\pi}{2}\right]$.		3				
302	03	Verify Cauchy's mean value theorem for $f(x) = \sqrt{x+1}, g(x) = 3x+2, x \in [3,8]$		3				
303	03	Evaluate: $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$	0.5	1	0	1	0.5	Doesn't exist
304	03	Evaluate: $\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\cos x}{x - \frac{\pi}{2}}\right)$	-1	1	0	1	-1	$\frac{\pi}{2}$

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305	03	Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\sin 2x}{x} \right)$	2	1	2	1	-1	$\frac{1}{2}$
306	03	Evaluate: $\lim_{x \rightarrow 2} \frac{x^2+x-6}{x^2-4}$	$\frac{5}{4}$	1	2	$\frac{5}{4}$	0	-1
307	03	Evaluate: $\lim_{x \rightarrow \infty} x^{\frac{1}{x}}$	1	1	∞	$-\infty$	1	0
308	03	Evaluate: $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n$	e	1	1	-1	e	$\frac{1}{e}$
309	03	Evaluate: $\lim_{x \rightarrow 0} x^x$	1	1	1	-1	e	$\frac{1}{e}$
310	03	Evaluate: $\lim_{x \rightarrow 0} (\cos x)^{\cot x}$	1	1	0	1	∞	None of these
311	03	Evaluate: $\lim_{x \rightarrow \pi} \frac{\sin x}{\pi - x}$	1	1	0	1	π	-1
312	03	Evaluate: $\lim_{n \rightarrow \infty} \left(\frac{n+4}{n} \right)^n$	e^4	1	1	e	e^3	e^4
313	03	The value of $\lim_{n \rightarrow \infty} \frac{1-n^2}{\sum n} = \text{---}$	-2	1	1	2	-2	Does not exist
314	03	Evaluate: $\lim_{x \rightarrow 0} \frac{\tan \pi x}{x}$	π	1	$\frac{1}{\pi}$	0	∞	π
315	03	The value of $\lim_{x \rightarrow \infty} x^2 e^{-x}$ is	0	1	1	e^1	0	e^2
316	03	Evaluate: $\lim_{x \rightarrow \frac{1}{2}} \frac{\cos^2 \pi x}{e^{2x} - 2ex}$		2				
317	03	Evaluate: $\lim_{x \rightarrow 0} \frac{x e^x - \log(1+x)}{x^2}$		2				
318	03	Evaluate: $\lim_{x \rightarrow a} \frac{\log(e^x - e^a)}{\log(x - a)}$		2				
319	03	Evaluate: $\lim_{x \rightarrow 1} (1 - x) \tan\left(\frac{\pi x}{2}\right)$		2				
320	03	Evaluate: $\lim_{x \rightarrow \frac{\pi}{2}} \cos x \log \tan x$		2				
321	03	Evaluate: $\lim_{x \rightarrow 0} \log x$		2				
322	03	Evaluate: $\lim_{x \rightarrow 1} \left(\frac{x}{x-1} - \frac{1}{\log x} \right)$		2				
323	03	Evaluate: $\lim_{x \rightarrow 0} \left[\frac{1}{x^2} - \frac{1}{\sin^2 x} \right]$		2				
324	03	Evaluate: $\lim_{x \rightarrow \frac{\pi}{2}} (\sec x - \tan x)$		2				
325	03	Evaluate: $\lim_{x \rightarrow 0} \left(1 - \frac{x}{2} \right)^{\frac{3}{x}} = ?$		2				
326	03	Evaluate: $\lim_{x \rightarrow 0} (a^x + x)^{\frac{1}{x}}$		3				
327	03	Evaluate: $\lim_{x \rightarrow \frac{\pi}{2}} (1 - \cos x)^{\tan x}$		3				
328	03	Evaluate: $\lim_{x \rightarrow 0} (\cot x)^{\sin x}$		3				
329	03	Evaluate: $\lim_{x \rightarrow 0} \left(\frac{1}{x} \right)^{1 - \cos x}$		3				
330	03	Evaluate $\lim_{x \rightarrow 0} \left(\frac{1^x + 2^x + 3^x}{3} \right)^{\frac{1}{x}}$		3				

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331	03	Evaluate: $\lim_{x \rightarrow \frac{\pi}{2}} (\sec x)^{\cot x}$		3				
332	03	Evaluate $\lim_{x \rightarrow 0} (\frac{e^x + e^{2x} + e^{3x}}{3})^{\frac{1}{x}}$		3				
333	03	Evaluate: $\lim_{x \rightarrow \frac{\pi}{2}} (\cos x)^{\frac{\pi}{2} - x}$		3				
334	03	Evaluate: $\lim_{x \rightarrow 0} \frac{\tan^2 x - x^2}{x^2 \tan^2 x}$		3				
335	03	Find: $\lim_{x \rightarrow 0} \frac{(1+x)^{\frac{1}{x}} - e}{x}$.		3				
336	03	If $\lim_{x \rightarrow 0} \frac{x(a + b \cos x) - c \sin x}{x^5} = 1$. Then find a,b and c.		3				
337	03	Find: $\lim_{x \rightarrow \frac{\pi}{2}} (\cos x)^{\frac{\pi}{2} - x}$.		3				
338	03	Find : $\lim_{x \rightarrow 0} \frac{x^2 + 2 \cos x - 2}{x \sin^3 x}$.		3				
339	03	Find: $\lim_{x \rightarrow \infty} \frac{x^n}{e^{kx}}$ ($k > 0$).		3				
340	04	Evaluate $(x, y) \xrightarrow{\lim} (0, 0) \frac{x - y + 2\sqrt{x} - 2\sqrt{y}}{\sqrt{x} - \sqrt{y}}$	2	01				
341	04	Evaluate $\lim_{(x,y) \rightarrow (1,2)} (x^2 y^3 - x^3 y^2 + 3x + 2y)$	11	01				
342	04	Find $\lim_{(x,y) \rightarrow (5,-2)} (x^4 + 4x^3 y - 5xy^2)$	-475	01				
343	04	Find the limit of $\lim_{(x,y) \rightarrow (0,0)} \cos \frac{x^2 + y^2}{x + y + 1}$	1	01				
344	04	$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - xy}{x + y} = \underline{\hspace{2cm}}$.	0	01	2	1	0	-1
345	04	$\lim_{(x,y) \rightarrow (0,1)} \frac{y^2 \tan^{-1} x}{x} = \underline{\hspace{2cm}}$	1	01	0	1	-1	∞
346	04	Value of the $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}}$ is	0	01	1	0	-1	Limit does not exist
347	04	$\lim_{(x,y) \rightarrow (0,0)} \frac{x - y}{x + y} = \underline{\hspace{2cm}}$.	Doesn't exist	01	0	-1	1	Doesn't exist
348	04	The value of $\lim_{(x,y) \rightarrow (1,1)} \frac{x - y}{x^2 - y^2}$	$\frac{1}{2}$	01	2	0	∞	$\frac{1}{2}$
349	04	If $= r \cos \theta$, $y = r \sin \theta$, then which is correct?	$r = \sqrt{x^2 + y^2}$, $\theta = \tan^{-1} \frac{y}{x}$	01	$r = x^2 + y^2, \theta = \frac{x}{y}$	$r = \sqrt{x^2 + y^2}, \theta = \tan\left(\frac{y}{x}\right)$	$r = x^2 + y^2, \theta = \tan^{-1} \frac{y}{x}$	$r = \sqrt{x^2 + y^2}, \theta = \tan^{-1} \frac{y}{x}$
350	04	Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$ OR If $f(x, y) = \frac{x^2 y}{x^4 + y^2}$ does $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exist?		02				
351	04	By considering different paths of approach, show that the function $f(x, y) = \frac{x^4 - y^2}{x^4 + y^2}$ has no limit as $(x, y) \rightarrow (0, 0)$		03				
352	04	Determine whether $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$ exist or not? If they exist find the value of the limit.		03				

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353	04	$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^3 - y^3}{x^2 + y^2}$ is		02				
354	04	Show that the function $f(x, y) = \frac{2x^2y}{x^4 + y^2}$ has no limit as (x, y) approaches to $(0, 0)$.		03				
355	04	Find $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{y^2 - x^2}$		03				
356	04	Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$ does not exists.		03				
357	04	Does limit of function $\frac{2xy}{x^2 + 3y^2}$ exist at $x = 0, y = 0$		04				
358	04	Determine whether $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y^2}{x^4 + 3y^4}$ exists or not? If it exist, find the value.		03				
359	04	Show that $f(x, y) = \begin{cases} \frac{2xy}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$ is continuous at every point except at the origin. OR Check the continuity for the following function at $(0, 0)$ $f(x, y) = \begin{cases} \frac{2xy}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$		03				
360	04	Discuss the continuity of the given function $f(x, y) = \begin{cases} \frac{x^3 - y^3}{x^2 + y^2}, & x \neq 0, y \neq 0 \\ 0, & x = 0, y = 0 \end{cases}$		03				
361	04	Discuss the continuity of the given function $f(x, y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$ OR Show that $f(x, y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$ is not continuous at $(0, 0)$.		04				
362	04	Show that the function $f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$ has no limit as (x, y) approaches to $(0, 0)$		04				
363	04	If $f(x, y) = \frac{y - x}{y + x}$ and $f(0, 0) = 0$, discuss the continuity of $f(x, y)$ at $(0, 0)$		03				
364	04	By considering different paths of approach, show that the function $f(x, y) = \frac{x^4 - y^2}{x^4 + y^2}$ has no limit as $(x, y) \rightarrow (0, 0)$		03				
365	04	Determine set of all point at which the function $f(x, y) = \frac{x^2 + y^2}{x - y}$ is continuous		03				
366	04	Show that $f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$ is continuous everywhere except at origin		03				
367	04	Show that $f(x, y) = \begin{cases} \frac{2x^2y}{x^3 + y^3}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$ is not continuous at the origin.		03				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
368	04	Show that $f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}}, & (x, y) \neq (0,0) \\ 0, & (x, y) = (0,0) \end{cases}$ is continuous at the origin.		04				
369	04	Discuss the continuity of $f(x, y) = \begin{cases} \frac{x}{\sqrt{x^2+y^2}}, & (x, y) \neq (0,0) \\ 2, & (x, y) = (0,0) \end{cases}$		04				
370	04	Determine the set of points at which the given function is continuous $f(x, y) = \begin{cases} \frac{3x^2y}{x^2+y^2}, & (x, y) \neq (0,0) \\ 0, & (x, y) = (0,0) \end{cases}$		04				
371	04	Determine the continuity of the function $f(x, y) = \begin{cases} (x^2 + y^2)\sin\left(\frac{1}{x^2+y^2}\right), & (x, y) \neq (0,0) \\ 0, & (x, y) = (0,0) \end{cases}$ at origin		03				
372	04	Let $f(x, y) = y \sin(xy)$.The value of $f_x(\pi, 1)$ is	-1	01	0	1	-1	2.5
373	04	If $z = x^2 - y^2$ then $\frac{\partial z}{\partial x} =$ _____.	none of these	01	2y	2z	0	none of these
374	04	If $f(x, y, z, w) = 3 \frac{\cos(xw) \sin y^5}{e^y + (1+y^2)/xyw} + 5xzw$, then $\frac{\partial f}{\partial z}$ at (1,2,3,4) is	20	01	20	200	0	1
375	04	If $x = r \cos \theta, y = r \sin \theta$ then the value of $\frac{\partial x}{\partial r}$	$\cos \theta$	01	$\cos \theta$	$\sec \theta$	$\operatorname{cosec} \theta$	$\sin \theta$
376	04	If $f = \log(x \tan^{-1} y)$ then f_{xy} is equal to _____.	0	01	0	1	$\frac{x}{1+y^2}$	$\frac{1}{x \tan^{-1} y}$
377	04	If $u = (x - y)^4 + (y - z)^4 + (z - x)^4$ then $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$ is	0	01	u	$4u$	$-u$	-1
378	04	If $f(x, y) = x^2y + xy^2$ then find $f_x(1,2)$ and $f_y(1,2)$ by definition		04				
379	04	Let $w = f(x, y, z)$ be a function of three independent variables, write the formal definition of the partial derivative for $\frac{\partial f}{\partial z}$ at (x_0, y_0, z_0) .		02				
380	04	Using this definition find $\frac{\partial f}{\partial z}$ at (1,2,3) for $f(x, y, z) = x^2yz^2$.		03				
381	04	Find the value $\frac{\partial f}{\partial y}$ at the point $(4, -5)$ for $f(x, y) = x^{2014} + 3xy + y - 2014$		02				
382	04	Find the values of $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at the point $(4, -5)$ if $f(x, y) = x^2 + 3xy + y - 1$		02				
383	04	Find the values of $\frac{\partial^2 f}{\partial x^2}$ and $\frac{\partial^2 f}{\partial x \partial y}$ at the point $(1, 2)$ for $f(x, y) = x^2 + 3xy + y - 1$		02				
384	04	If $f(x) = x^3 + x^2y^3 - 2y^2$, find $f_x(2,1)$ and $f_y(2,1)$		04				
385	04	If $x = r \cos \theta$ and $y = r \sin \theta$, find $\frac{\partial x}{\partial r}$ and $\frac{\partial r}{\partial x}$		03				

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386	04	If $u = x^2y + y^2z + z^2x$ then find out $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$		03				
387	04	If $u = f(e^{y-z}, e^{z-x}, e^{x-y})$, show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.		03				
388	04	If $u = \sin(x^2 + y^2)$ and $a^2x^2 + b^2y^2 = c^2$, find $\frac{du}{dx}$.		03				
389	04	If $u = x^2 + y^2 + z^2$, where $x = e^t, y = e^t \sin t, z = e^t \cos t$, find $\frac{du}{dt}$ at $t = 1$.		03				
390	04	If $z = f(x, y), x = uv, y = \frac{u+v}{u-v}$, prove that $u \frac{\partial z}{\partial u} + v \frac{\partial z}{\partial v} = 2x \frac{\partial z}{\partial x}$.		03				
391	04	If $z = f(x, y), x = uv, y = \frac{u+v}{u-v}$, Prove that $u \frac{\partial z}{\partial u} + v \frac{\partial z}{\partial v} = 2x \frac{\partial z}{\partial x}$		03				
392	04	If $f(u, v), u = \log(x^2 + y^2), v = \frac{y}{x}$, show that $x \frac{\partial z}{\partial y} - y \frac{\partial z}{\partial x} = (1 + v^2) \frac{\partial z}{\partial v}$.		03				
393	04	If $u = x \log(xy)$, where $x^3 + y^3 + 3xy = 1$, find $\frac{du}{dx}$		03				
394	04	If resistors of R_1, R_2 and R_3 ohms are connected in parallel to make an R -ohm resistor, the value of R can be found from the equation $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$. Find the value of $\frac{\partial R}{\partial R_2}$ when $R_1 = 30, R_2 = 45$ and $R_3 = 90$ ohms		03				
395	04	The amount of work done by the heart's main pumping chamber, the left ventricle, is given by the equation $W = PV + \frac{\rho \delta v^2}{2g}$, where W is the work per unit time, P is the average blood pressure, V is the volume of blood pumped out during the unit of time, δ is the weight density of the blood, v is the average velocity of the existing blood, and g is the acceleration of the gravity which is constant. Find partial derivative of W with respect to each variable		04				
396	04	If $u = \frac{e^{x+y+z}}{e^x + e^y + e^z}$ then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 2u$		03				
397	04	If $u = \log(\tan x + \tan y + \tan z)$ then prove that $\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2$		04				
398	04	If $x^3 + y^3 = 6xy$ then find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$		04				
399	04	Suppose that $x^2 + y^2 = r^2$ and $x = r \cos \theta$ as in polar coordinates. Find $\left(\frac{\partial x}{\partial r}\right)_\theta$ and $\left(\frac{\partial r}{\partial x}\right)_y$		04				
400	04	Verify that the function $u = e^{-\alpha^2 k^2 t} \cdot \sin kx$ is a solution of the heat conduction equation $u_t = \alpha^2 u_{xx}$.		04				
401	04	For the function $u = (x^2 + y^2)^{\frac{1}{5}}$ the value of $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy}$ is	-6u/25	01	-6u/25	-u/8	6u/25	-1/5u

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402	04	$f(x,y) = \frac{\cos(xy+x^3y)}{x+x^3}$ find f_{xy} at (0,1).	-1	01	0	-2	2	-1
403	04	If $u = f(x + ay) + g(x - ay)$ then $\frac{\partial^2 u}{\partial y^2}$ is	$a^2 \frac{\partial^2 u}{\partial x^2}$	01	$a \frac{\partial^2 u}{\partial x^2}$	$-a^2 \frac{\partial^2 u}{\partial x^2}$	$-\frac{\partial^2 u}{\partial x^2}$	$a^2 \frac{\partial^2 u}{\partial x^2}$
404	04	If $u = (3xy - y^3) - (y^2 - 2x)^{\frac{3}{2}}$ then $u_{xy} =$ _____	$3 + \frac{3y}{\sqrt{y^2 - 2x}}$	01	$3 + \frac{3y}{\sqrt{y^2 - 2x}}$	$3 - \frac{3y}{\sqrt{y^2 - 2x}}$	$\frac{-3}{\sqrt{y^2 - 2x}}$	$3 + \frac{3}{\sqrt{y^2 + 2x}}$
405	04	Determine whether $u(x,y) = \ln\sqrt{x^2 + y^2}$ is a solution Laplace's equation		05				
406	04	Find all first and second order partial derivatives for $f(x,y) = x^2 \sin y + y^2 \cos x$ Hence, verify mixed derivative Theorem		05				
407	04	If $u = x^3y + e^{xy^2}$, then prove that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$		04				
408	04	If $z = x + y^x$, then prove that $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$		04				
409	04	If $u = \log(x^2 + y^2)$, verify that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$		04				
410	04	If $u = 2(ax + by)^2 - (x^2 + y^2)$ and $a^2 + b^2 = 1$ then prove that $\frac{d^2 u}{dx^2} + \frac{d^2 u}{dy^2} = 0$		04				
411	04	If $\theta = t^n e^{-\frac{r^2}{4t}}$ then find n so that $\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = \frac{\partial \theta}{\partial t}$		04				
412	04	If $u = \log(\tan x + \tan y)$ then prove that $\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} = 2$.		03				
413	04	If $u = \log(x^3 + y^3 - x^2y - xy^2)$ prove that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right)^2 u = -\frac{4}{(x+y)^2}$.		03				
414	04	If $u = x^2y + y^2z + z^2x$, find the values of $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$ and $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$		04				
415	04	If $u = x^2y + y^2z + z^2x$, prove that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u = 6(x + y + z)$		07				
416	04	If $u = x^2y + y^2z + z^2x$, prove that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u = 3(u_{xx} + u_{yy} + u_{zz})$		07				
417	04	Compute the four second order partial derivatives of $f(x,y) = xy^2 + 3x^2e^y$.		04				
418	04	If $u = e^{3xyz}$ show that $\frac{\partial^3 u}{\partial x \partial y \partial z} = (3 + 27xyz + 27x^2y^2z^2)e^{3xyz}$		04				
419	04	For $u = \tan^{-1}\left(\frac{y}{x}\right)$. Show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ and $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$		04				
420	04	Show that $\frac{\partial^2 z}{\partial x \partial y} = -[x \log(ex)]^{-1}$ at the point $x = y = z$ for the surface $x^x y^y z^z = c$		07				
421	04	If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, then prove that (a) $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$ (b) $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u = \frac{-9}{(x+y+z)^2}$		07				
422	04	If $z(x + y) = x^2 + y^2$, show that $\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} \right)^2 = 4 \left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} \right)$		04				

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423	04	Show that $\frac{\partial^2 \Omega}{\partial x \partial y} = \frac{\partial^2 \Omega}{\partial y \partial x}$ where $\Omega = y + x^y$		03				
424	04	If $u = e^{xyz}$, show that $\frac{\partial^3 u}{\partial x \partial y \partial z} = (1 + 3xyz + x^2 y^2 z^2) e^{xyz}$		03				
425	04	If $u = f(x, y)$, where $x = e^s \cos t$ and $y = e^s \sin t$, show that $\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = e^{-2s} \left[\left(\frac{\partial u}{\partial s}\right)^2 + \left(\frac{\partial u}{\partial t}\right)^2\right]$		03				
426	04	Find the value of n so that $v = r^n (3 \cos^2 \theta - 1)$ satisfy the equation $\frac{\partial}{\partial r} \left(r^2 \frac{\partial v}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial v}{\partial \theta} \right) = 0$		03				
427	04	Find the value of n for which $u = t^n e^{-\frac{r^2}{4kt}}$ satisfies the partial differential equation $\frac{\partial u}{\partial t} = k \left(\frac{\partial^2 u}{\partial r^2} + \frac{2}{r} \frac{\partial u}{\partial r} \right)$.		03				
428	04	If $u = (x^2 + y^2 + z^2)^m$ then find $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$		03				
429	04	If $u = (1 - 2xy + y^2)^{-\frac{1}{2}}$ then find the value of $\frac{\partial}{\partial x} \left[(1 - x^2) \frac{\partial u}{\partial x} \right] + \frac{\partial}{\partial y} \left[y^2 \frac{\partial u}{\partial y} \right]$		03				
430	04	If $u = f(r^2)$, where $r^2 = x^2 + y^2 + z^2$ then prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 4r^2 f''(r^2) + 6f'(r^2)$		03				
431	04	Verify that the function $z = \ln(e^x + e^y)$ is solution of the differential equations $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 1$ and $\frac{\partial^2 z}{\partial x^2} \frac{\partial^2 z}{\partial y^2} - \left(\frac{\partial^2 z}{\partial x \partial y} \right)^2 = 0$		03				
432	04	If $V = (1 - 2xy + y^2)^{-\frac{1}{2}}$, then show that $\frac{\partial}{\partial x} \left[(1 - x^2) \frac{\partial V}{\partial x} \right] + \frac{\partial}{\partial y} \left[y^2 \frac{\partial V}{\partial y} \right] = 0$		03				
433	04	For $u = \log \left(\frac{x^3 + y^3}{x + y} \right)$, find $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy}$		03				
434	04	If $u = \sin^{-1} \frac{x}{y} + \tan^{-1} \frac{y}{x}$ then the value of $xu_x + yu_y$ is	0	01	u	$-u$	0	1
435	04	If $f(x, y) = \frac{x}{y} + \sin \left(\frac{y}{x} \right) e^{y/x}$, then the value of $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y}$ is	0	01	1	0	-1	f
436	04	If $u = x^3 \cos \left(\frac{y}{x} \right)$ then $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y}$ is _____	$3u$	01	u	$2u$	$3u$	none of these
437	04	If $u = y^2 f \left(\frac{x}{y} \right)$ then $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} =$ _____	$2u$	01	$6u$	0	u	$2u$
438	04	$u = \sin^{-1} \left(\frac{x}{y} \right)$ is a homogeneous function of degree	0	01	$\frac{1}{2}$	0	1	-1
439	04	Which of the following is homogeneous function of degree one?	$\frac{x^2}{y}$	01	$\frac{x^2}{y}$	$\frac{x^2}{y^2}$	$\frac{x+2}{y}$	$\frac{yx}{yx+xy}$

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440	04	If $f = e^{\frac{x}{y}} \tan\left(\frac{x}{y}\right) + e^{\frac{y}{x}} \cot\left(\frac{y}{x}\right)$, then value of $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y}$ is	0	01	0	f	$-f$	$2f$
441	04	The function $f(x, y) = x^2 y f\left(\frac{y}{x}\right)$ is homogeneous of degree is -----.	3	01	0	1	2	3
442	04	If $u = y \tan^{-1}\left(\frac{x}{y}\right) + x \cot^{-1}\left(\frac{y}{x}\right)$ then $xu_x + yu_y =$ _____	u	01	0	u	$2u$	$3u$
443	04	If $u = \tan^{-1}(\log x - \log y)$ then $xu_x + yu_y =$ _____	0	01	u	0	$-u$	$2u$
444	04	Which of the following is homogeneous function of degree one?	$\frac{x^2 + y^2}{x + y}$	01	$\ln x - \ln y$	$x^2 y + x^3$	$\tan(x^2 + y^2)$	$\frac{x^2 + y^2}{x + y}$
445	04	If $u = x^3 \cos\left(\frac{y}{x}\right)$ then $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y}$ is _____	$3u$	01	$3u$	$-3u$	u	$2u$
446	04	If $u = \tan^{-1}[\log(x^2 y) - \log(xy^2)]$ then $xu_x + yu_y =$ _____	0	01	u	0	$3u$	e^u
447	04	If $u = \tan^{-1}\left(\frac{x^2 + y^2}{x - y}\right)$, then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \sin 2u$		03				
448	04	If $u = \sin^{-1}\left(\frac{\frac{1}{x^4} + \frac{1}{y^4}}{\frac{1}{x^5} + \frac{1}{y^5}}\right)$, then find the value of $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$ and $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$		05				
449	04	State Euler's theorem on homogeneous function. Using it show that (i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$ (ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2 \sin u \cos 3u$. If $u = \tan^{-1}(x^2 + 2y^2)$ Explain whether we can apply Euler's theorem for the function $u = f(x, y) = \frac{x^2 + y^2 + 1}{x + y}$		05				
450	04	If $u = \tan^{-1}(x^2 + 2y^2)$ then show that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2 \sin u \cos 3u$		04				
451	04	If $u = \tan^{-1}\left(\frac{x^3 + y^3}{x - y}\right)$, show that $xu_x + yu_y = \sin 2u$. Also prove that $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = 2 \cos 3u \sin u$.		05				
452	04	If $u = e^{x^2 + y^2 - xy}$ then prove that (i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u \ln u$ (ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2u \ln u (2 \ln u + 1)$		05				
453	04	Give the example of the non-homogenous function in x and y . If $u = \tan^{-1}(x^2 + xy + 2015y^2)$ then show that i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$ & ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2 \sin u \cos 3u$		07				
454	04	If $u(x, y) = \sin^{-1}\left(\frac{x + y}{\sqrt{x} + \sqrt{y}}\right)$ then prove that $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = \frac{-\sin(u) \cos(2u)}{4 \cos^3 u}$		05				

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455	04	If $u(x,y) = \log\left(\frac{x^2+y^2}{x+y}\right)$ then show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 1$		03				
456	04	If $u = \cos^{-1}\left(\frac{x^3+y^3}{x+y}\right)$ show that $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{\sin 2u \sin u - 4 \cos u}{\sin^3 u}$		05				
457	04	State Euler's theorem. If $z = x^n f\left(\frac{y}{x}\right) + y^n f\left(\frac{x}{y}\right)$ then prove that $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} + x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n^2 z$		05				
458	04	If $u = \operatorname{cosec}^{-1}\left(\frac{x+y}{x^2+y^2}\right)$, then show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan u$		03				
459	04	State Euler's theorem of homogeneous function of two variables and apply it to evaluate $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ for $u = \frac{x^3 y^3}{x^3 + y^3}$		05				
460	04	Verify Euler's theorem when $u = \frac{\frac{1}{x^4} + \frac{1}{y^4}}{\frac{1}{x^5} + \frac{1}{y^5}}$		05				
461	04	If $u = \sin^{-1}\left(\frac{x+2y}{x^3+y^3}\right)$, then find the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$		03				
462	04	Let $u = f(x,y)$ is a homogeneous function of degree n . Then find $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$	nu	01				
463	04	If $u = xy^2 + y^3 + x^3 + z^3$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 3u$		03				
464	04	If $u = \frac{\frac{1}{x^4} + \frac{1}{y^4}}{\frac{1}{x^5} + \frac{1}{y^5}}$ then find values of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ and $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2}$		05				
465	04	If $u = \ln\left(\frac{x^7+y^7+z^7}{x+y+z}\right)$ then show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 6$		03				
466	04	State modified Euler's theorem and find the degree of homogeneous function $u(x,y) = \frac{1}{x^2} + \frac{1}{xy} + \frac{(\log x - \log y)}{x^2}$		03				
467	04	If $u = x^3 y^2 \sin^{-1}\left(\frac{y}{x}\right)$, show that (i) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 5u$ (ii) $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 20u$		04				
468	04	If $u = \operatorname{cosec}^{-1}\left(\frac{x^{\frac{1}{2}}+y^{\frac{1}{2}}}{\frac{1}{x^3}+\frac{1}{y^3}}\right)^{\frac{1}{2}}$, then show that $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{144}(13 + \tan^2 u)$. State modified Euler's theorem.		07				
469	04	If $u = \sin^{-1}\left(\frac{\frac{1}{x^4}+\frac{1}{y^4}}{\frac{1}{x^6}+\frac{1}{y^6}}\right)$ then show that $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{144}(\tan^2 u - 11)$.		04				
470	04	If $u = \sin^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, then show that		05				

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		$(1)2x \frac{\partial u}{\partial x} + 2y \frac{\partial u}{\partial y} = \tan u$ $(2)x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{1}{4}(\tan^3 u - \tan u)$						
471	04	State Euler's theorem on homogeneous function of two variables. If $u = \tan^{-1}\left(\frac{x^2+y^2}{x+y}\right)$, prove that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -2\sin^3 u \cos u$.		05				
472	04	State modified Euler's Theorem. Show that, if $u = \tan^{-1}\left(\frac{x^2+y^2}{x-y}\right)$ then $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = \frac{1}{4} \sin 4u - \frac{1}{2} \sin 2u$		05				
473	04	For $u = \log(x^3 - x^2 y + xy^2 - y^3)$, find $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy}$		04				
474	04	State Euler's theorem for homogeneous function and verify it for $u = \sqrt{x} + \sqrt{y} + \sqrt{z}$ by direct differentiation		07				
475	04	Verify Euler's theorem for $f(x, y) = \frac{(x^3+y^3)}{x+y}$		07				
476	04	If $u = \tan^{-1}\left(\frac{x^3+y^3}{x+y}\right)$ then prove that (1) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$ (2) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \sin 2u(1 - 4 \sin^2 u)$		07				
477	04	If $u = \sin^{-1}\left(\frac{x^2+y^2}{\frac{3}{x^2+y^2}}\right)$ then show that (i) $xu_x + yu_y = \frac{1}{2} \tan u$ (ii) $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = \frac{1}{4}(\tan^3 u - \tan u)$		07				
478	04	Verify Euler's theorem when $u = \frac{x^{\frac{1}{3}}+y^{\frac{1}{3}}}{x^{\frac{1}{5}}+y^{\frac{1}{5}}}$		07				
479	04	If $u = \frac{1}{x^7} \sin^{-1}\left(\frac{x^2+y^2}{x^2+2xy}\right)$ then find the value of $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} + x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ at $x = 1, y = 2$.		03				
480	04	If $u = y^2 - 4ax, x = at^2, y = 2at$, then $\frac{du}{dt} = \underline{\hspace{2cm}}$	0	01	$16a^2 t$	$-16a^2 t$	$16at$	$-16at$
481	04	Find $\frac{dw}{dt}$ if $w = xy + z, x = \cos t, y = \sin t, z = t$.		02				
482	04	If $z = e^{xy}, x = t \cos t, y = t \sin t$ Find $\frac{dz}{dt}$ at $t = \frac{\pi}{2}$		03				
483	04	If $r^2 = x^2 + y^2 + z^2$ and $v = r^m$, prove that $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = m(m+1)r^{m-2}$.		07				
484	04	If $u = (x^2 + y^2 + z^2)^m$ then find $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$		07				
485	04	If $u = r^m$ then prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = m(m+1)r^{m-2}$ Where $r^2 = x^2 + y^2 + z^2$		07				
486	04	If $u = x^2 y^3, x = \log t, y = e^t$ find $\frac{du}{dt}$.		03				

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487	04	For $z = \tan^{-1}\left(\frac{x}{y}\right)$, $x = u \cos v$, $y = u \sin v$, evaluate $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$ at the point $\left(1.3, \frac{\pi}{6}\right)$		03				
488	04	If $u = x^2 + y^2$, $x = a \cos t$, $y = b \sin t$ then find $\frac{du}{dt}$		03				
489	04	If $u = f(r)$ where $r^2 = x^2 + y^2$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r}f'(r)$		05				
490	04	If $u = f(r)$ where $x = r \cos \theta$, $y = r \sin \theta$ Prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r}f'(r)$		05				
491	04	If $u = f(r)$ and $r^2 = x^2 + y^2 + z^2$ then show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = f''(r) + \frac{2}{r}f'(r)$		05				
492	04	If $u = x^4 y + y^2 z^3$ where $x = r s e^t$, $y = r s^2 e^{-t}$, $z = r^2 s \sin t$ Find the value of $\frac{\partial u}{\partial s}$ when $r = 1, s = 1, t = 0$		04				
493	04	Suppose that $u = f(x, y, z)$ and $x = g_1(t)$, $y = g_2(t)$, $z = g_3(t)$. Then write the chain rule for derivative of u w.r.t. t . Express $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$ in terms of r and s . If $w = x + 2y + z^2$, $x = \frac{r}{s}$, $y = r^2 + \log(s)$, $z = 2r$		05				
494	04	If $u = f(x - y, y - z, z - x)$, then find $u_x + u_y + u_z$		04				
495	04	If $u = f(x^2 + 2yz, y^2 + 2zx)$, prove that $(y^2 - zx)\frac{\partial u}{\partial x} + (x^2 - yz)\frac{\partial u}{\partial y} + (z^2 - xy)\frac{\partial u}{\partial z} = 0$.		04				
496	04	If $f(lx + my + nz, x^2 + y^2 + z^2) = 0$, prove that $(ly - mx) + (ny - mz)\frac{\partial z}{\partial x} + (lz - nx)\frac{\partial z}{\partial y} = 0$.		04				
497	04	If $u = f(l, m)$ where $l = x + y$ and $m = x - y$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 2\frac{\partial u}{\partial l}$.		04				
498	04	If $u = f(x, y)$, where $x = e^s \cos t$ and $y = e^s \sin t$, show that $\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = e^{-2s} \left[\left(\frac{\partial u}{\partial s}\right)^2 + \left(\frac{\partial u}{\partial t}\right)^2\right]$.		04				
499	04	If $z = f(x, y)$ where $x = s + t$, $y = s - t$ Show that $\left(\frac{\partial z}{\partial x}\right)^2 - \left(\frac{\partial z}{\partial y}\right)^2 = \frac{\partial z}{\partial s} \cdot \frac{\partial z}{\partial t}$		04				
500	04	If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$		07				
501	04	If $u = f(r, s, t)$ and $r = \frac{x}{y}$, $s = \frac{y}{z}$, $t = \frac{z}{x}$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$		07				
502	04	If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$ show that $x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} + z^2\frac{\partial u}{\partial z} = 0$		07				

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503	04	By using partial derivatives find the value of $\frac{dy}{dx}$ for $xe^y + \sin xy + y - \log 2 = 0$ at $(0, \log 2)$		02																			
504	04	Find $\frac{dy}{dx}$ when $y^{x^y} = \sin x$		03																			
505	04	If $u = \tan^{-1}\left(\frac{x}{y}\right)$ where $x^2 + y^2 = a^2$ Find $\frac{du}{dx}$		04																			
506	04	Using partial derivatives, find $\frac{d^2y}{dx^2}$ for $x^5 + y^5 = 5x^2$.		04																			
507	04	Find $\frac{dw}{dt}$ at $t = 0$ if $w = x y^2 + z^2$, where $x = \cos t, y = \sin t, z = t$		03																			
508	04	If $u = \sin^{-1}(x - y), x = 3t, y = 4t^3$ then show that $\frac{du}{dt} = \frac{3}{\sqrt{1-t^2}}$		03																			
509	04	If $z = f(x, y), x = e^u + e^{-v}, y = e^{-u} - e^v$ then prove that $\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}$		04																			
510	04	If $z = x^2y + 3xy^4$ where $x = \sin 2t$ and $y = \cos t$ Find $\frac{dz}{dt}$ when $t=0$		03																			
511	04	If $f(x, y) = 0, \phi(y, z) = 0$ Prove that $\frac{\partial f}{\partial y} \frac{\partial \phi}{\partial z} \cdot \frac{dz}{dx} = \frac{\partial f}{\partial x} \frac{\partial \phi}{\partial y}$		03																			
512	04	If $u = x^2 - y^2$, where $x = 2r - 3s + 4, y = -r + 8s - 5$, find $\frac{\partial u}{\partial r}$ and $\frac{\partial u}{\partial s}$		03																			
513	04	If $u = f(2x - 3y, 3y - 4z, 4z - 2x)$, Prove that $\frac{1}{2} \frac{\partial u}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial y} + \frac{1}{4} \frac{\partial u}{\partial z} = 0$		07																			
514	04	If $u = f(xy)$ show that $x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} = 0$		03																			
515	04	If $z = xy^2 + x^2y, x = at^2, y = 2at$, find $\frac{dz}{dt}$		03																			
516	04	Suppose f is a differentiable function of x and y and $g(u, v) = f(e^u + \sin v, e^u + \cos v)$. Use the following table to calculate $g_u(0,0), g_v(0,0)$. <table><tr><td></td><td>f</td><td>g</td><td>f_x</td><td>f_y</td></tr><tr><td>(0,0)</td><td>3</td><td>6</td><td>4</td><td>8</td></tr><tr><td>(1,2)</td><td>6</td><td>3</td><td>2</td><td>5</td></tr></table>		f	g	f_x	f_y	(0,0)	3	6	4	8	(1,2)	6	3	2	5		04				
	f	g	f_x	f_y																			
(0,0)	3	6	4	8																			
(1,2)	6	3	2	5																			
517	04	For an implicit function $f(x, y) = c$ the value of $\frac{dy}{dx}$ is	$-\frac{f_x}{f_y}$	01	$\frac{f_x}{f_y}$	$\frac{f_y}{f_x}$	$-\frac{f_x}{f_y}$	$-\frac{f_y}{f_x}$															
518	04	For the function $u = (x^2 + y^2)^{\frac{1}{3}}$ the value of $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$ is	$-\frac{2u}{9}$	01	$-2u$	$-\frac{2u}{3}$	$\frac{u}{9}$	$-\frac{2u}{9}$															
519	04	Value of $\frac{dy}{dx}$ for $ax^2 + 2hxy + by^2 = 1$ is _____	$-\frac{ax + hy}{hx + by}$	01	$\frac{hx + by}{ax + hy}$	$\frac{ax + hy}{hx + by}$	$-\frac{ax + hy}{hx + by}$	$-\frac{hx + by}{ax + hy}$															
520	04	Using partial differentiation, find $\frac{dy}{dx}$ for $(\cos x)^y = x^{\sin y}$		03																			

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521	04	Find $\frac{dy}{dx}$ when $(\cos x)^y = (\sin y)^x$		04				
522	04	If $x^3 + y^3 - 3axy = 0$ find $\frac{dy}{dx}$		03				
523	04	If $x^2 + y^2 - xy = 0$, find $\frac{dy}{dx}$		03				
524	05	Equation of tangent plane of $z = x$ at (2,0,2) is	$z = x$	01	$z = x$	$+y + z = 2$	$x + z = 0$	None of these
525	05	Equation of tangent plane to the surface $z = x^2 - y^2$ at point (5, -4,9) is	$(z - 9) = 10(x - 5) + 8(y + 4)$	01	$(z - 9) = 10(x - 5) + 8(y + 4)$	$(z - 9) = 10(x - 5) - 8(y - 4)$	$(z - 9) = 8(x - 5) + 10(y + 4)$	$(z - 9) = 8(x - 5) - 10(y + 4)$
526	05	Equation of tangent plane to the surface $x + y + z = 0$ at point (1,1,1) is	$x + y + z = 3$	01	$x - y + z = 3$	$x + y + z = 3$	$x + y - z = 3$	None of these
527	05	Equation of normal line to the surface $x + y + z = 0$ at point (0,0,0) is	$x = y = z$	01	$x = -y = z$	$x = y = -z$	$x = y = z$	None of these
528	05	Equation of Tangent plane to the surface $z = 2x^2 + y^2$ at the point (1, 1, 3)	$4x + 2y - z = 3$	01	$2x + 2y - z = 3$	$4x + 2y - z = 3$	$4x + 2y - 14z = 3$	$11x + 2y - z = 3$
529	05	Equation of tangent plane to the surface $z + 8 = xe^y \cos z$ at point (8,0,0) is _____	$x + 8y - z - 8 = 0$	01	$x + 8y - z - 8 = 0$	$x + 8y + z + 8 = 0$	$-x + 8y - z + 8 = 0$	$2x + 8y - 8z - 8 = 0$
530	05	Equation of tangent plane of $x - y = 0$ at (0,0,2) is	$x = y$	01	$x + y = 0$	$x + y = 2$	$x = y$	$x = y + 2$
531	05	Find the equation of tangent plane and the normal line to the surface $2x^2 + y^2 + 2z = 3$ at (2,1, -3).		04				
532	05	Find the equation of tangent plane and normal line of the surface $f(x, y, z) = x^2 + y^2 + z - 9 = 0$ at the point P(1,2,4).		04				
533	05	Find the equation of tangent plane and normal line to the surface $\cos \pi x - x^2 y + e^{xz} + yz = 4$ at the point P(0,1,2).		03				
534	05	Find the equation of tangent plane and normal line of the surface $x^2 + y^2 - z^2 = 4$ at the point (1,2,1).		03				
535	05	Find the equation of tangent Plane to the elliptic paraboloid $z = 2x^2 + y^2$ at the point (1,1,3)		04				
536	05	Find the equation of tangent plane and normal line to the surface $x^2 yz + 3y^2 = 2xz^2 - 8z$ at the point (1,2, -1).		04				
537	05	Find the equation of the tangent plane and the normal line of the surface $x^2 + 2y^2 + 3z^2 = 12$ at the point (1,2, -1).		04				
538	05	Find the equation of tangent plane and normal line to the surface $xyz = 6$ at (1,2,3)		04				
539	05	Check that the surfaces $z = xy - 2$ and $x^2 + y^2 + z^2 = 3$ have the same tangent plane at (1,1, -1) or not		03				
540	05	Find the equation of the tangent plane and normal line to the surface $x^3 + 2xy^2 - 7z^2 + 3y + 1 = 0$ at point (1,1,1).		04				
541	05	Find the equation of tangent plane and normal line at a point (3, 4,5) to the surface $x^2 + y^2 - 4z = 5$		03				

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542	05	Find the equation of the tangent plane and normal line to the surface $\frac{x^2}{2} - \frac{y^2}{3} = z$ at the point $(2, 3, -1)$.		03				
543	05	Find the equation of the tangent plane and normal line to the surface $2xz^2 - 3xy - 4x = 7$ at $(1, -1, 2)$		04				
544	05	Find the tangent plane and normal line at the point P_o on the given surface $z^2 - 2x^2 - 2y^2 - 12 = 0$, $P_o(1, -1, 4)$		04				
545	05	Find the equation of the tangent plane and normal line to the following surfaces at a given point 1) $\sqrt{x} + \sqrt{y} + \sqrt{z} = 4$ at $(4, 1, 1)$. 2) $z = \tan^{-1}(\frac{y}{x})$ at $(1, 1)$.		04				
546	05	Find the equation of the tangent plane and normal line to the surface $z + 7 = xe^y \cos z$ at the point $(7, 0, 0)$.		04				
547	05	Find the equation of tangent plane and normal line at the point $(-2, 1, -3)$ to the ellipsoid $\frac{x^2}{4} + y^2 + \frac{z^2}{9} = 3$		04				
548	05	Find the equations of tangent plane and normal line of the surface $x^2 + y^2 + z^2 = 3$ at the point $(1, -1, 1)$.		04				
549	05	Find the equation of the tangent plane and normal line to the surface $x^3 + 2xy^2 - 7z^2 + 3y + 1 = 0$ at point $(1, 2, 3)$.		04				
550	05	If $f(x, y) = e^{xy^2}$, the total differential of the function at the point $(1, 2)$ is	$4e^4(dx + dy)$	01	$e(dx + dy)$	$-e(dx + dy)$	$4e^4(dx + dy)$	$e^4(4dx + dy)$
551	05	The base radius and height of a right circular cone are measured as 10 cm and 25 cm, respectively, with a possible error in measurement of as much as 0.1 cm in each. Use differentials to estimate the maximum error in the calculated volume of the cone.		04				
552	05	Use differentials to approximate the increase in the surface area of a soap bubble when its radius increases from 4 inches to 4.1 inches.		04				
553	05	The height of a cylinder is measured as 12 cm with a possible error of ± 0.1 cm. Evaluate the volume of the cylinder with radius 4 cm and give an estimate for the possible error in this value.		04				
554	05	If $z = x^2 + 3xy - y^2$ then find the differential dz if x changes from 2 to 2.05 and y changes from 3 to 2.96.		04				
555	05	A point (a, b) is said to be a saddle point if at (a, b)	$rt - s^2 < 0$	01	$rt - s^2 > 0$	$rt - s^2 = 0$	$rt - s^2 < 0$	$rt - s^2 \geq 0$
556	05	The minimum value of $f(x, y) = x^2 + y^2$ is	0	01	1	2	4	0
557	05	A point (a, b) is said to be an extreme point if at (a, b)	$rt - s^2 > 0$	01	$rt - s^2 > 0$	$rt - s^2 = 0$	$rt - s^2 < 0$	$rt - s^2 \leq 0$
558	05	The stationary point of the function $f(x, y) = x^2 + xy + y^2$ is	$(0, 0)$	01	$(1, 0)$	$(1, 1)$	$(0, 0)$	$(0, -1)$
559	05	If $z = x^2 + y^2 + 3$ minimum value of $z =$ _____.	3	01	3	0	∞	None of these
560	05	The stationary points of $x^4 + y^4 - 2x^2 + 4xy - 2y^2$ are	$(\sqrt{2}, -\sqrt{2})$	01	$(\sqrt{2}, -\sqrt{2})$	$(\sqrt{2}, \sqrt{2})$	$(-\sqrt{2}, -\sqrt{2})$	$(-\sqrt{2}, 0)$

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561	05	The stationary points of $f(x, y) = xy + (x - y)$ are_____	$(1, -1)$	01	(0,0)	$(1, -1)$	(1,2)	$(1, -2)$
562	05	The function $z = 5xy - 4x^2 + y^2 - 2x - y + 5$ has _____ at $x = \frac{1}{41}, y = \frac{18}{41}$.	saddle point	01	maxima	saddle point	minima	maxima & minima
563	05	For what values of the constant k does the second derivative test guarantee that $f(x, y) = x^2 + kxy + y^2$ will have a saddle point at (0,0)? A minimum at (0,0)?		04				
564	05	Find the maximum and minimum values and saddle points of $f(x, y) = x^4 + y^4 - 4xy + 1$		05				
565	05	Find maxima and minima of the function $f(x, y) = x^3 + y^3 - 3x - 12y + 20$.		05				
566	05	Find the maximum and minimum of the function $x^3 + y^3 - 63(x + y) + 12xy$		07				
567	05	Find all the maxima minima and saddle points of the function $f(x, y) = x^2 - 2xy + 2y^2 - 2x + 2y + 1$		04				
568	05	Find the extreme values of $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$		07				
569	05	Examine $f(x, y) = x^3 + y^3 - 3axy$ for maximum and minimum values		04				
570	05	Find the extreme values for $x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$		04				
571	05	Find the extreme values of the function $x^3 + 3xy^2 - 3x^2 - 3y^2 + 7$		04				
572	05	Find a point within a triangle such that the sum of the squares of its distances from the three vertices is a minimum.		05				
573	05	Find the extreme values of $\sin x + \sin y + \sin(x + y), (0 \leq x, y \leq \frac{\pi}{2})$.		05				
574	05	Find the local extreme values of $f(x, y) = xy - x^2 - y^2 - 2x - 2y + 4$.		07				
575	05	Find the extreme values of the function $f(x, y) = x^3 + y^3 - 3xy$		03				
576	05	Find the points of maxima and minima of $x + y + \frac{1}{x} + \frac{1}{y}$. Also find the maximum and minimum values of the function.		04				
577	05	In a plane triangle ABC find the extreme values of $\cos A \cos B \cos C$.		04				
578	05	Find the maximum and minimum values and saddle point, if any of $f(x, y) = x^2 + y^2 + 2x + 4y + 9$		04				
579	05	Find the maximum and minimum values $f(x, y) = 2(x^2 + y^2) - x^4 + y^4$		04				
580	05	Discuss the Maxima and Minima of the function $3x^2 - y^2 + x^3$		04				
581	05	Discuss the maxima and minima of the function $x^2 + y^2 + 6x + 12$		03				
582	05	Find the maximum and minimum values of $f(x, y, z) = x - 2y + 5z$ on the sphere $x^2 + y^2 + z^2 = 30$.		03				
583	05	Find the maximum and minimum value of $f(x, y) = y^2 - y \cos x, -1 \leq x \leq 7$		03				
584	05	Find the extreme values of the function $x^3 + xy^2 + 21x - 12x^2 - 2y^2$.		03				
585	05	Find Extreme values of $f(x, y) = y^3 + 3x^2y - 6x^2 - 6y^2 + 2$		03				

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586	05	Find the extreme values of $\sin x . \sin y . \sin(x + y)$		03				
587	05	Find extreme values of $x^4 + y^4 - 2x^2 + 4xy - 2y^2$.		03				
588	05	Divide 24 into three parts such that the continued product of the first, square of the second and cube of the third may be maximum.		03				
589	05	Find the extreme values of the function $x^3 + xy^2 + 21x - 12x^2 - 2y^2$		03				
590	05	Find the extreme values of $x^2y^2 - 5x^2 - 8xy - 5y^2$		03				
591	05	The length l of the rectangle area increases at 1.5 ft/sec and its breadth b increases at 0.5ft/sec. Find the rate of increase of area of the rectangle when l=4 ft and b= 3ft		03				
592	05	Show that if perimeter of triangle is a constant, the triangle has maximum area when it is equilateral.		03				
593	05	The base of an aquarium with a given volume V is made of slate and the sides are made of glass. If slate costs five times as much (per unit area) as glass, find the dimensions of the aquarium that minimize the cost of the materials		03				
594	05	A closed rectangular box has length twice its breadth and has constant volume V, Determine the dimensions of the box requiring least surface area.		04				
595	05	Find the minimum distance from the point $\left(0,0,\frac{25}{9}\right)$ to the surface $z = xy$.		03				
596	05	Find the minimum distance from origin to the plane $3x + 2y + z = 12$		03				
597	05	Find the point on the plane $x + 2y + 3z = 13$ closest to the point (1,1,1).		07				
598	05	Find the greatest and smallest values that the function $f(x,y) = xy$ takes on the ellipse $\frac{x^2}{8} + \frac{y^2}{2} = 1$		05				
599	05	Find the maximum and minimum values of $x + y$ on the circle $x^2 + y^2 = 4$		07				
600	05	The pressure P at any point (x,y,z) in the space is $P = 400xyz^2$. Find the highest pressure at the surface of the unit sphere $x^2 + y^2 + z^2 = 1$		05				
601	05	The temperature T at any point (x,y,z) in the space is $P = 400xyz^2$. Find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$		05				
602	05	Find the maximum and minimum values of the function $f(x,y) = 3x + 4y$ on the circle $x^2 + y^2 = 1$ using the method of Lagrange multipliers.		03				
603	05	Use Lagrange method to find the shortest distance from the point (1,2,2) to the sphere $x^2 + y^2 + z^2 = 16$		07				
604	05	Using Lagrange's method of undetermined multipliers, find the maximum value of $u = x^p y^q z^r$ when the variable x,y,z are subject		05				

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		to the condition $ax + by + cz = p + q + r$						
605	05	Find the points on the sphere $x^2 + y^2 + z^2 = 4$ that are closest to and farthest from the point $(3,1,-1)$		07				
606	05	Find a point on the plane $2x + 3y - z = 5$ which is nearest to the origin, using Lagrange's method.		05				
607	05	Find two positive numbers whose product is 100 and sum is minimum		05				
608	05	Find the minimum value of x^2yz^3 subject to the condition $2x + y + 3z = a$ using Lagrange's method		05				
609	05	If the surface area of a right circular cylinder is given then prove that its height is equal to the diameter of its base when the volume is maximum.		04				
610	05	Find the maximum and minimum distance from the point $(1,2,2)$ to the sphere $x^2 + y^2 + z^2 = 36$.		07				
611	05	Find two non-negative numbers whose sum is 9 such that the product of one number and the square of the other is maximum.		03				
612	05	Find the minimum value of $x^2 + y^2 + z^2$, given that $ax + by + cz = p$		04				
613	05	Find the numbers $x, y, \text{ and } z$ such that $xyz = 8$ and $xy + yz + zx$ is maximum, using the Lagrange's method		05				
614	05	Find the shortest and longest distance from the point $(1,2,-1)$ to the sphere $x^2 + y^2 + z^2 = 24$		07				
615	05	Show that the rectangular solid of maximum volume that can be inscribed in a sphere is cube.		04				
616	05	Use Lagrange's method of multipliers to find the maximum value of $f(x, y, z) = xyz$ if it is given that $x + y + z = 24$		05				
617	05	A rectangular box without a lid is to be made from $12m^2$ of cardboard. Find the maximum volume of such a box		07				
618	05	A rectangular box open at the top is to have a volume of 32 cubic units. Find the dimensions of the box requiring least material for its construction		07				
619	05	You are planning to make an open rectangular box from 8 -in.-by-15-in. piece of cardboard by cutting congruent squares from the corners and folding up the sides. What are the dimensions of the box of largest volume you can make this way, and what is its volume?		07				
620	05	Find the numbers $x, y \text{ and } z$ such that $xyz = 16$ and $18xy + 16yz + 12xz$ is minimum, using the Lagrange's method.		07				
621	05	A rectangular box open from the top is to have a volume of 108 c.c. find the dimension of the box requiring least material for its construction.		07				
622	05	Use the method of the Lagrange's multipliers to find the volume of the largest rectangular		07				

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		parallelepiped which is 8xyz that can be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.						
623	05	If $x = u + 3v, y = v - u$ then value of $J = \frac{\partial(x,y)}{\partial(u,v)}$ is	4	01	-1	4	5	7
624	05	The Jacobian of polar coordinates (r, θ) with respect to Cartesian coordinates (x, y) is	None of these	01	r	$r \sin \theta$	r^2	None of these
625	05	If $x = r \cos \theta, y = r \sin \theta$ then $\frac{\partial(r,\theta)}{\partial(x,y)}$ is	$\frac{1}{r}$	01	r	$-r$	$\frac{1}{r}$	$-\frac{1}{r}$
626	05	If $J = \frac{\partial(x,y)}{\partial(r,\theta)}$ and $J' = \frac{\partial(r,\theta)}{\partial(x,y)}$ then the value of JJ' is	1	01	0	1	-1	Always $J = J'$
627	05	If $J = \frac{\partial(u,v)}{\partial(x,y)} = 3$ then $J' = \frac{\partial(x,y)}{\partial(u,v)} =$ _____	$\frac{1}{3}$	01	3	1	$\frac{1}{3}$	None of these
628	05	If $x = u(1 - v), y = uv$ the value of $\frac{\partial(u,v)}{\partial(x,y)} =$ _____.	$\frac{1}{u}$	01	$\frac{1}{u}$	$\frac{1}{v}$	u	0
629	05	If $x = u + 3v, y = v - u$ then value of $J = \frac{\partial(x,y)}{\partial(u,v)}$ is _____	4	01	5	-1	4	1
630	05	If $x = u(1 + v)$ and $y = v(1 + u)$ then $\frac{\partial(x,y)}{\partial(u,v)} =$ _____	$1 + u + v$	01	1	u	v	$1 + u + v$
631	05	The transformation from $r\theta z$ – space to xyz – space is given by the equations $x = r \cos \theta, y = r \sin \theta, z = z$. Find the Jacobian $J(r, \theta, z)$.		03				
632	05	Find the Jacobian of the transformation, $x = \rho \sin \phi \cos \theta, y = \rho \sin \phi \sin \theta, z = \rho \cos \phi$		03				
633	05	If $x = \sqrt{vw}, y = \sqrt{wu}, z = \sqrt{uv}$ and $u = r \sin \theta \cos \phi, v = r \sin \theta \sin \phi, w = r \cos \theta$. Calculate $\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)}$		04				
634	05	Find the Jacobian of the transformation $x = u + 4v$ and $y = 3u - 2v$.		04				
635	05	Find $\frac{\partial(u,v)}{\partial(x,y)}$, where $u = x^2 - y^2, v = 2xy$		04				
636	05	Solve the system $u = 2x + y, v = x - 2y$ for x and y in terms of u and v . Also find $\frac{\partial(x,y)}{\partial(u,v)}$		03				
637	05	If $u = 2xy, v = x^2 - y^2, x = r \cos \theta, y = r \sin \theta$. Calculate $\frac{\partial(u,v)}{\partial(r,\theta)}$		03				
638	05	If $x = e^v \sec u$ and $y = e^v \tan u$ then in usual notations find the values of J and J'		04				
639	05	If $x = r \cos \theta$ and $y = r \sin \theta$ by finding J and J' separately, show that $JJ' = 1$		04				
640	05	If $u = x + y$ and $v = \frac{x}{x+y}$, find $\frac{\partial(u,v)}{\partial(x,y)}$		03				
641	05	Find Jacobian $\frac{\partial(u,v)}{\partial(x,y)}$ for functions $u = xsiny, v = ysinx$		03				
642	05	Find the Jacobian $\frac{\partial(u,v)}{\partial(x,y)}$ for $u = \frac{x+y}{1-xy}, v = \tan^{-1} x + \tan^{-1} y$		03				
643	06	Find: $\int \cos 5x \, dx$	$\frac{\sin 5x}{5} + c$	1	$5 \sin 5x + c$	$-5 \sin x + c$	$\frac{\cos 5x}{5} + c$	$\frac{\sin 5x}{5} + c$
644	06	Find: $\int \sec^2 x \, dx$	$\tan x + c$	1	$\tan x + c$	$\cot x + c$	$-\tan x$	$2 \sec^2 x + c$
645	06	$\int \frac{1}{x^2 + 4x + 13} \, dx =$ _____	$\frac{1}{3} \tan^{-1} \left(\frac{x+2}{3} \right) + c$	1	$\tan^{-1} \left(\frac{x+2}{3} \right) + c$	$\frac{1}{3} \tan^{-1} \left(\frac{x+3}{2} \right)$	$\frac{1}{3} \tan^{-1} \left(\frac{x+2}{3} \right)$	$\tan^{-1} \left(\frac{x+3}{2} \right)$

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646	06	$\int \frac{dx}{1 + \cos x} = \text{————}$	$\tan \frac{x}{2} + c$	1	$\frac{1}{2} \tan \frac{x}{2} + c$	$\tan \frac{x}{2} + c$	$2 \tan \frac{x}{2} + c$	$\tan^2 \frac{x}{2} + c$
647	06	Find: $\int \cos x \, dx$	$\sin x + c$	1	$\cos x + c$	$\sin x + c$	$\tan x + c$	$\operatorname{cosec} x + c$
648	06	Find: $\int \tan x \, dx$	$\log \sec x + c$	1	$\log \sec x + c$	$\log \cos x + c$	$\log \sin x + c$	$\log \tan x + c$
649	06	Find: $\int \sin x \, dx$	$-\cos x + c$	1	$-\sin x + c$	$-\sec x + c$	$-\cos x + c$	$\cos x + c$
650	06	Find: $\int \cot x \, dx$	$\log \sin x + c$	1	$\log \tan x + c$	$\log \sec x + c$	$\log \cos x + c$	$\log \sin x + c$
651	06	Find: $\int \sqrt{(3x - 1)(x + 2)} \, dx.$		3				
652	06	Find: $\int (2x - 1)^2 \cdot x^{-1} \, dx$		2				
653	06	Find: $\int \sqrt{25 - x^2} \, dx.$		2				
654	06	Find: $\int 3 \sec^7 x \tan x \, dx .$		2				
655	06	Find: $\int \frac{2x^3 + 20x - 1}{x^2 + 10} \, dx$		2				
656	06	Find: $\int \sqrt{x^2 + 9} \, dx$		2				
657	06	Find: $\int \frac{1}{x^2 + 4x + 9} \, dx$		2				
658	06	Find: $\int \frac{x}{(x-1)(x-2)} \, dx$		2				
659	06	Find: $\int \frac{1}{\sqrt{9x^2 - 16}} \, dx$		2				
660	06	Find: $\int \frac{1}{x^2 + 5x + 6} \, dx$		2				
661	06	Find: $\int \frac{1}{\sqrt{16x^2 + 1}} \, dx$		2				
662	06	Find: $\int \frac{1}{x^2 - x - 6} \, dx$		2				
663	06	Find: $\int 4^{5x+2} \, dx$		2				
664	06	Find: $\int \sqrt{4x^2 + 4x - 15} \, dx$		2				
665	06	Find: $\int \frac{1}{\sqrt{4x^2 - 4x - 5}} \, dx$		2				
666	06	Solve: $\int e^{(5x-3)} \, dx$		2				
667	06	Find: $\int \frac{1}{1 - \cos 2x} \, dx$		2				
668	06	Find: $\int \left(\frac{x}{a} + \frac{a}{x} + x^a + a^x + 5 \right) dx$		2				
669	06	Find: $\int \frac{x^3}{x^2 - 9} \, dx$		2				
670	06	Find: $\int \frac{x^2 - 10}{x^2 + 9} \, dx$		2				
671	06	Find: $\int \frac{1}{\sqrt{8 - 2x - x^2}} \, dx$		2				
672	06	Find: $\int \frac{1}{\sqrt{25x^2 - 1}} \, dx$		2				
673	06	Find: $\int \sqrt{(x - 4)(x + 4)} \, dx$		2				
674	06	Find: $\int \frac{1}{3x^2 + 5} \, dx$		2				
675	06	Find: $\int \frac{x^3}{x^2 - 16} \, dx$		2				
676	06	Find: $\int 7^{9x-2} \, dx$		2				
677	06	Solve: $\int \sqrt{x^2 - 4} \, dx$		2				
678	06	Solve: $\int 3^{5x+2} \, dx$		2				
679	06	Find: $\int \frac{x^3}{(x-2)(x+3)} \, dx$		1.5				
680	06	Find: $\int e^{3x} \cos 2x \, dx$		2				
681	06	Solve : $\int \cos 4x \cos 2x \sin x \, dx$		2				
682	06	Find: $\int \sqrt{16 - x^2} \, dx.$		2				

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683	06	Evaluate: $\int \sqrt{(1 - \sin 2x)} \, dx$		2				
684	06	Evaluate: $\int \frac{\sec^2 x}{\sqrt{\tan^2 x + 4}} \, dx$		2				
685	06	Evaluate: $\int \frac{2x+3}{x^3-3x+2} \, dx$		2				
686	06	The value of $\int \log x (\log x + 2) \, dx =$ _____.	$x (\log x)^2 + c$	1	$x (\log x)^2 + c$	$x (1 + \log x)^2 + c$	$x (\log x)^3 + c$	$x [1 + (\log x)^2] + c$
687	06	Find: $\int x^2 \log x \, dx$	$\frac{x^3}{3} \log x - \frac{x^3}{9} + c$	1	$\frac{x^3}{3} \log x + \frac{x^3}{9} + c$	$\frac{x^3}{6} \log x - \frac{x^3}{9} + c$	$\frac{x^3}{3} \log x - \frac{x^3}{9} + c$	$\frac{x^3}{6} \log x - \frac{x^4}{9} + c$
688	06	Find: $\int x^3 \cos 2x \, dx$		2				
689	06	Find: $\int x^2 e^x \, dx$		2				
690	06	Find: $\int x^2 \sin x \, dx$		2				
691	06	Find: $\int x^2 \sin 2x \, dx$		2				
692	06	Find: $\int x \sin x \, dx$		2				
693	06	Evaluate: $\int e^{2x} \sin(3x) \, dx$		2				
694	06	Find: $\int x^2 \cos 7x \, dx$		2				
695	06	Evaluate: $\int (1 + \sqrt{x^2 + 1}) 2x \, dx$		2				
696	06	Find: $\int (2x - 1)^2 \cos nx \, dx$		2				
697	06	The value of $\iint 3y \, dx \, dy$ over the triangle the vertices $(-1,1)$, $(0,0)$ and $(1,1)$	2	1	0	1	2	3
698	06	$\int_1^2 \int_1^2 \frac{1}{xy} \, dx \, dy =$ _____	$(\log 2)^2$	1	0	$(\log 2)^2$	1	$\log 2$
699	06	What does the region of $\int_1^4 \int_2^6 \, dx \, dy$ represent?	Rectangle	1	Rectangle	Square	Circle	Triangle
700	06	$\int_0^2 \int_0^{x^2} e^{\frac{y}{x}} \, dy \, dx$ is equal to _____	$e^2 - 1$	1	$e^2 - 1$	e^2	$e^2 + 1$	e^{-2}
701	06	$\int_0^1 \int_0^y e^{x/y} \, dx \, dy =$ _____	$\frac{e - 1}{2}$	1	$\frac{e - 1}{2}$	$e - 1$	e	$+1$
702	06	The values of $\int_0^1 \int_0^1 (3x^2 - 2y^2) \, dx \, dy$ is -----	1/3	1	0	1	-1	1/3
703	06	The value of $\int_1^2 \int_0^x \frac{\ln x}{x} \, dy \, dx =$ ____.	$2 \ln 2 - 1$	1	$2 \ln 2 - 2$	$2 \ln 2 - 1$	$\ln 2$	0
704	06	The region $\int_{-1}^2 \int_1^4 \, dx \, dy$ represents ____.	rectangle	1	rectangle	Line	circle	triangle
705	06	What does the region of $\int_2^5 \int_3^7 \, dx \, dy$ represent?	rectangle	1	Rectangle	Circle	Square	cardiod
706	06	Evaluate: $\int_1^2 \int_{-1}^1 (1 + 3xy) \, dx \, dy$	2	1	$\frac{1}{4}$	2	$\frac{1}{6}$	0
707	06	$\int_0^1 \int_x^{x^2} xy \, dy \, dx =$ ____.	-1/24	1	1/24	1/12	-1/24	-1/12
708	06	Evaluate $\int_0^1 \int_0^{\sqrt{x}} (x^2 + y^2) \, dy \, dx$		3				
709	06	Evaluate $\int_0^1 \int_0^1 \frac{dx \, dy}{(\sqrt{1-x^2} \sqrt{1-y^2})}$		2				
710	06	Evaluate: $\int_{-1}^1 \int_0^2 (1 - 6x^2 y) \, dx \, dy$.		3				
711	06	Evaluate the integral $\int_0^1 \int_0^{\sqrt{1-y^2}} (x^2 + y^2) \, dx \, dy$		3				

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712	06	Evaluate: $\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dydx}{(1+x^2+y^2)}$		2				
713	06	Evaluate the integral $\int_1^2 \int_y^2 xy \, dx \, dy$.		4				
714	06	Evaluate: $\int_1^2 \int_0^1 (1 + 3xy) \, dx \, dy$.		5				
715	06	Evaluate $\int_0^1 \int_0^x (x^2 + y^2) \, dA$, where dA indicates small area in $xy - plane$.		2				
716	06	Evaluate: $\int_0^1 \int_0^x e^{\frac{y}{x}} \, dy \, dx$		3				
717	06	Evaluate $\int_0^1 \int_y^{1+y^2} x^2 y \, dx \, dy$		4				
718	06	Evaluate $\int_0^1 \int_{x^2}^x (1 + xy) \, dy \, dx$		3				
719	06	Evaluate: $\int_0^4 \int_0^{x^2} e^{y/x} \, dy \, dx$		3				
720	06	Evaluate: $\int_{\frac{\pi}{2}}^{\pi} \int_1^2 x \cos xy \, dy \, dx$		3				
721	06	Evaluate: $\int_0^1 \int_1^2 xy \, dy \, dx$		3				
722	06	Evaluate: $\int_1^4 \int_{2x^2}^{3x^2} x e^{x^2+y} \, dy \, dx$		3				
723	06	Evaluate: $\int_0^3 \int_0^1 (x^2 + 3y^2) \, dy \, dx$.		4				
724	06	Evaluate: $\int_3^6 \int_0^1 (1 + 6xy) \, dx \, dy$		4				
725	06	Evaluate: $\int_3^6 \int_0^1 (1 + 8x^2y) \, dx \, dy$		4				
726	06	Evaluate $\iint_R (x + y) \, dy \, dx$ R is the region bounded by $x = 0, x = 2, y = x, y = x + 2$		4				
727	06	Evaluate $\iint_R xy \, dA$, where R is the region bounded by x – axis, ordinate $x = 2a$ and the curve $x^2 = 4ay$		4				
728	06	Evaluate $\iint_R e^{2x+3y} \, dA$ where R is triangle bounded by $x = 0, y = 0, x + y = 1$.		4				
729	06	Evaluate $\int \int (x^2 + y^2) \, dx \, dy$ over the region bounded by the lines $y = 4x, x + y = 3, y = 0, y = 2$		7				
730	06	Sketch the region of integration and evaluate it: $\int_1^4 \int_0^{\sqrt{x}} \frac{3}{2} e^{y/\sqrt{x}} \, dy \, dx$		3				
731	06	Evaluate $\iint_R (y - 2x^2) \, dA$ where R is the region bounded by $ x + y = 1$.		5				
732	06	Evaluate $\int \int (xy) \, dx \, dy$ where the region R is bounded by $y^2 + x^2 - 2x = 0, y^2 = 2x, y = x$		4				
733	06	Evaluate $\iint_R (2x - y^2) \, dA$; over the triangular region R enclosed between the lines $y = -x + 1, y = x + 1$ & $y = 3$.		3				
734	06	Evaluate $\iint_R (x^2) \, dA$, where R is the region in the first quadrant bounded by the hyperbola $xy = 16$ and the lines $y = x, y = 0$ and $x = 8$.		7				
735	06	Integrate $f(x, y) = x^2 + y^2$ over the triangular region with vertices $(0,0), (1,0),$ and $(0,1)$.		5				
736	06	Sketch the triangle R in the $xy - plane$ bounded by the $x - axis$, the line $y = 2x$, and the line $x = 1$. Evaluate $\iint_R \frac{\sin x}{x} \, dA$		3				
737	06	Evaluate $\iint_R x \, dA$, where R is the region bounded by the parabolas $y^2 = 4x$ and $x^2 = 4y$.		3				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
738	06	Evaluate $\int \int y \, dx dy$ over the region bounded by in the first quadrant. $y = x^2, x = 0, x + y = 2$		5				
739	06	Evaluate $\int \int y \, dx dy$ where A is the region bounded by the parabolas $4x = y^2, x^2 = 4y$		4				
740	06	Evaluate $\iint_D (x + 2y) dA$, where D is the region bounded by the parabolas $y = 2x^2$ and $y = x + 1$.		4				
741	06	Evaluate $\iint_S \sqrt{xy - y^2} \, dx dy$ Where S is a triangle with vertices $(0, 0), (10, 1), (1, 1)$.		3				
742	06	Evaluate $\iint_R xy(x + y) dA$ over the area between $y = x^2$ and $y = x$.		4				
743	06	Evaluate $\iint_R x y dA$ over the area between $y = x^2$ and $y = x$		4				
744	06	Evaluate $\iint (6x^2 + 2y) dx dy$ over the region R bounded between $y = x^2$ and $y = 4$.		4				
745	06	Evaluate $\iint_R y \sin(xy) \, dA$, where R is the region bounded by $x = 1, x = 2, y = 0$ and $y = \frac{\pi}{2}$.		3				
746	06	Evaluate $\int_R xy \, dy dx$, where R is the positive quadrant of the circle $x^2 + y^2 = a^2$.		7				
747	06	Evaluate $\int \int (x^2 - y^2) dx dy$ over the triangle with vertices $(0,1), (1,1), (1,2)$		4				
748	06	Find $\iint_R y \, dA$, where R is the region enclosed by the parabola $y = x^2$ and the line $y = x + 2$.		4				
749	06	Change the order of integration and then evaluate: $\int_0^{\frac{a}{\sqrt{2}}} \int_x^{\sqrt{a^2 - x^2}} y^2 \, dA$		4				
750	06	Evaluate the integral $\int_0^2 \int_{\frac{y}{2}}^1 e^{x^2} \, dx dy$ by changing the order of integration.		4				
751	06	Evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} \, dA$ by changing the order of integration.		2				
752	06	Evaluate the iterated integral $\int_0^1 \int_x^1 \sin(y^2) \, dy dx$.		3				
753	06	Evaluate $\int_0^1 \int_{x^2}^{2-x} xy \, dA$ by changing the order of integration.		4				
754	06	Change the order of integration and evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \frac{e^y}{(e^y+1)\sqrt{1-x^2-y^2}} \, dy dx$.		7				
755	06	Sketch the region of integration, reverse the order of integration and evaluate the integral $\int_0^2 \int_0^{4-x^2} \frac{x e^{2y}}{4-y} \, dy dx$		4				
756	06	Evaluate: $\int_0^2 \int_{y-2}^{2y} xy dx dy$ by changing the order of integration.		4				
757	06	Change the order of integration and then evaluate: $\int_0^\infty \int_0^x x e^{\frac{-x^2}{y}} \, dy dx$		7				
758	06	Sketch the region of integration. Also Change the order of integration and evaluate the integral $\int_0^1 \int_y^{\sqrt{y}} dx dy$		5				
759	06	Evaluate: $\int_0^a \int_{a-\sqrt{a^2-y^2}}^{a+\sqrt{a^2-y^2}} dx dy$ By changing the order of integration.		4				

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760	06	Evaluate: $\int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dydx$ By changing the order of integration.		4				
761	06	Change the order of Integrations and hence evaluate: $\int_0^1 \int_{2x}^2 e^{y^2} dydx$.		4				
762	06	Sketch the region of integration and evaluate by reversing the order of integration. $\int_0^{\frac{3}{2}} \int_3^{6-2x} x dydx$		3				
763	06	Evaluate: $\int_0^3 \int_1^{\sqrt{4-y}} (x+y) dx dy$ By changing the order of integration.		4				
764	06	Evaluate: $\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x}{\sqrt{(x^2+y^2)}} dydx$ by changing the order of integration.		4				
765	06	Change the order of integration and evaluate $\int_0^a \int_y^a \frac{x}{x^2+y^2} dx dy$		4				
766	06	Evaluate: $\int_0^a \int_{\sqrt{ax}}^a \frac{y^2 dy dx}{\sqrt{y^4-a^2x^2}}$ by changing the order to integration.		7				
767	06	Evaluate: $\int_0^{2a} \int_{\frac{x^2}{4a}}^{3a-x} (x^2+y^2) dA$ by changing the order of integration.		7				
768	06	Evaluate: $\int_0^\infty \int_{-y}^y (y^2-x^2) e^{-y} dx dy$ by changing the order of integration.		4				
769	06	By changing the order of the integration, evaluate: $\int_0^3 \int_y^3 \frac{x}{x^2+y^2} dx dy$.		4				
770	06	Evaluate the integral by the changing the order of integration. $\int_0^8 \int_{\sqrt[3]{y}}^2 \sqrt{x^4+1} dx dy$		4				
771	06	Sketch the region of integration, reverse the order of integration and evaluate the integral $\int_0^{\sqrt{3}} \int_0^{3-x^2} \frac{x e^y}{3-y} dy dx$		4				
772	06	Evaluate $\int_0^{\frac{1}{2}} \int_0^{\sqrt{1-4y^2}} \frac{1+x^2}{\sqrt{1-x^2} \sqrt{1-x^2-y^2}} dx dy$, using change of order.		4				
773	06	Change the order of integration and evaluate $\int_0^1 \int_0^{\sqrt{1-y^2}} \frac{\cos^{-1} x}{\sqrt{1-x^2-y^2} \sqrt{1-x^2}} dx dy$		5				
774	06	Evaluate by changing the order of the integral $\int_{-2}^4 \int_{\frac{y^2}{2}-3}^{y+1} xy dA$		5				
775	06	Sketch the region of integration. Also Change the order of integration and evaluate the integral $\int_0^1 \int_x^{\sqrt{x}} dy dx$		5				
776	06	$\int_0^{\frac{\pi}{4}} \int_0^1 r dr d\theta = \underline{\hspace{2cm}}$	$\frac{\pi}{8}$	1	$\pi/4$	$\frac{\pi}{8}$	$\frac{\pi}{2}$	π
777	06	The curve $r = 2$ is _____	circle with center origin and radius 2	1	straight line	point at distance '2' on initial line	circle with center origin and radius 2	cardioid
778	06	$\int_0^\pi \int_0^{\sin \theta} r dr d\theta = \underline{\hspace{2cm}}$	$\frac{\pi}{4}$		$\frac{\pi}{2}$	$\frac{\pi}{4}$	π	$-\frac{\pi}{2}$
779	06	$\int_0^{\frac{\pi}{2}} \int_0^{2 \cos \theta} r^5 dr d\theta = \underline{\hspace{2cm}}$	$\frac{5\pi}{3}$		$\frac{5\pi}{3}$	$\frac{10\pi}{3}$	$\frac{32\pi}{3}$	$\frac{2\pi}{3}$
780	06	Evaluate the integral $\int_0^{\pi/2} \int_0^{1-\sin \theta} r^2 \cos \theta dr d\theta$		2				
781	06	Evaluate: $\int_0^\infty e^{-x^2} dx$		3				
782	06	Evaluate: $\int_0^\pi \int_0^{a(1+\cos \theta)} r dr d\theta$		3				

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783	06	Evaluate: $\int_0^\pi \int_0^{(1-\cos\theta)} r^2 \sin\theta \, dr d\theta$		2				
784	06	Evaluate $\iint_R r \sin\theta \, dr d\theta$ over the area bounded by the cardioid $r = a(1 + \cos\theta)$ above the initial line.		4				
785	06	Evaluate $\iint_R \sin\theta \, dA$, where R is the region in the first quadrant that is outside the circle $r = 2$ and inside the cardioids $r = 2(1 + \cos\theta)$.		5				
786	06	Evaluate $\int \int_R r^3 \sin 2\theta \, dr d\theta$ over the area bounded in the first quadrant between the circles $r = 2$ and $r = 4$		3				
787	06	Evaluate $\iint r^3 \, dr d\theta$ over the area bounded between the circles $r = 2 \cos\theta$ and $r = 4 \cos\theta$.		3				
788	06	Evaluate $\iint r\sqrt{a^2 - r^2} \, dr d\theta$ over the upper half of the circle $r = a \cos\theta$.		4				
789	06	Evaluate: $\iint_R \frac{r \, dr d\theta}{\sqrt{a^2 + r^2}}$ Where R is a loop of $r^2 = a^2 \cos 2\theta$.		4				
790	06	Evaluate $\iint_A r \sin\theta \, dr d\theta$ over the area of the curve $r = \frac{(1+\cos\theta)}{2}$ above the initial line.		4				
791	06	Evaluate: $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} \, dx dy$.		4				
792	06	Integrate $f(x, y) = \frac{\ln(x^2+y^2)}{\sqrt{x^2+y^2}}$ over the region $1 \leq x^2 + y^2 \leq e$ by changing to polar coordinates.		4				
793	06	Change the Cartesian integral $\int_0^1 \int_0^{\sqrt{1-x^2}} e^{-(x^2+y^2)} \, dy dx$ into an equivalent polar integral. Then evaluate the same.		2				
794	06	Change to polar coordinates and then evaluate: $\int_0^a \int_y^a \frac{x}{x^2+y^2} \, dx dy$		7				
795	06	Evaluate the integral $\int_0^1 \int_0^x \sqrt{x^2 + y^2} \, dy dx$ By changing into the polar coordinates.		3				
796	06	Change into polar coordinates and hence. evaluate: $\int_0^a \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} y^2 \, dy dx$.		4				
797	06	Evaluate: $\int_0^1 \int_0^1 \, dx dy$ by changing to polar co-ordinates.		3				
798	06	Evaluate $\int_0^a \int_0^{\sqrt{a^2-y^2}} y^2 \sqrt{x^2 + y^2} \, dx dy$, by changing into polar coordinates.		3				
799	06	Change into polar co-ordinate and evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} e^{-(x^2+y^2)} \, dy dx$		3				
800	06	Evaluate the integral $\int_0^{4a} \int_{\frac{y^2}{4a}}^{\frac{y}{2} \frac{x^2-y^2}{x^2+y^2}} \, dx dy$ by transforming into Polar coordinates.		4				
801	06	Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} (x^2 + y^2) \, dy dx$ by changing into polar coordinates where $a > 0$		3				
802	06	Evaluate $\iint_R (x^2 + y^2) \, dx dy$ over the positive quadrant of the circle $x^2 + y^2 = a^2$ by polar coordinates.		4				
803	06	Use polar coordinates to evaluate $\iint_R (x + y) \, dx dy$ over the positive quadrant of the circle $x^2 + y^2 = a^2$		4				
804	06	Evaluate $\int_0^2 \int_0^{\sqrt{4-x^2}} (x^2 + y^2) \, dy dx$ by changing into the polar coordinates.		4				
805	06	Evaluate the integrals $\int_0^\infty \int_0^\infty \frac{1}{(1+x^2+y^2)^2} \, dx dy$ by changing into polar co-ordinates		3				

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806	06	Evaluate $\int_0^1 \int_x^{\sqrt{2-x^2}} (x^2 + y^2) dy dx$ by changing to polar coordinates.		5				
807	06	Evaluate: $\int_0^1 \int_x^{\sqrt{2x-x^2}} (x^2 + y^2) dy dx$ by changing into polar coordinates.		4				
808	06	If $u = x - y$ & $v = x + y$ then the value of the $J = \frac{\partial(x,y)}{\partial(u,v)}$ is _____	$\frac{1}{2}$	1	1	-1	1/2	1/4
809	06	Evaluate $\iint_R (x^2 + y^2) dA$ by changing the variables, where R is the region lying in the first quadrant and bounded by the hyperbolas $x^2 - y^2 = 1, x^2 - y^2 = 9, xy = 2$ and $xy = 4$.		4				
810	06	Evaluate $\int_0^4 \int_{\frac{y}{2}}^{\frac{y}{2}+1} \frac{2x-y}{2} dx dy$ by applying the transformations $u = \frac{2x-y}{2}, v = \frac{y}{2}$. Draw both the regions.		4				
811	06	Evaluate $\iint_E (y - x) dx dy$, over the region E in the XY-plane bounded by the straight lines $y = x - 3, y = x + 1, 3y + x = 5, 3y + x = 7$.		4				
812	06	Evaluate $\iint [(x^2 - y^2)]^2 dA$ over the area bounded by the lines $ x + y = 1$, using the transformations $x + y = u; x - y = v$.		4				
813	06	Evaluate $\iint [(x + y)]^2 dx dy$ where R is the parallelogram in the xy-plane with vertices (1,0), (3,1), (2,2), (0,1) using the transformation $u = x + y$ and $v = x - 2y$		4				
814	06	Evaluate the integral $\int_0^1 \int_0^{1-x} e^{\frac{y}{x+y}} dy dx$ by changing the variables $x + y = u, y = uv$		4				
815	06	Evaluate $\int_0^1 \int_0^{1-x} \sqrt{x+y} (y - 2x)^2 dy dx$ using change of variables $x + y = u$ and $y - 2x = v$		7				
816	06	Evaluate $\int_R x dx dy$, R is the triangle (0,0), (1,0) & (1,1) using transformations $x = u, y = uv$.		7				
817	06	Using the transformation $-y = u, x + y = v$, evaluate $\iint \cos\left(\frac{x-y}{x+y}\right) dx dy$ over the region bounded by vertices (0,0), (1,0) & (0,1).		3				
818	06	Evaluate $\iint_R (x^2 + y^2) dA$ by changing the variables, where R is the region lying in the first quadrant and bounded by the hyperbolas $x^2 - y^2 = 1, x^2 - y^2 = 2, xy = 2$ and $xy = 4$. Where $x^2 - y^2 = u, 2xy = v$.		5				
819	06	Evaluate $\iint_R (x + y) e^{x^2-y^2} dA$, Where R is the region bounded by lines $x - y = 0, x - y = 2, x + y = 0$ and $x + y = 3$ using transformation $x - y = u$ and $x + y = v$		5				
820	06	Evaluate $\iint_D \frac{y^2}{x} dx dy$, where D is the region between the parabolas $x = 1 - y^2$ and $x = 3(1 - y^2)$. Use the change of variables $(x, y) = (v(1 - u^2), u)$.		4				

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821	06	Using the transformation $x + y = u$ & $x - y = v$ evaluate $\iint_R \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) dA$, where R is the triangular region with vertices (0,0), (2,0), (1,1).		4				
822	06	$\int_0^1 \int_0^1 \int_0^1 e^{x+y+z} dx dy dz =$ ____	(e-1) ³	1	1	e ³	(e – 1) ³	(1 – e) ³
823	06	The value of $\int_0^1 \int_0^x \int_0^y dz dy dx$	$\frac{1}{6}$	1	1	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{6}$
824	06	$\int_0^2 \int_1^3 \int_1^2 xy^2z dz dy dx =$ _____	26		26	-26	25	1
825	06	Evaluate , $\iiint_D \sqrt{x^2 + y^2} dV$ where D is the solid bounded by the surfaces $x^2 + y^2 = z^2, z = 0, z = 1$.		3				
826	06	Evaluate: $\int_0^4 \int_0^4 \int_0^4 (1 + xyz) dx dy dz.$		3				
827	06	Evaluate: $\int_0^1 \int_0^{1-x} \int_0^{x+y} (e^z) dz dy dx.$		3				
828	06	Evaluate the integral $\int_0^2 \int_1^z \int_0^{yz} xyz dx dy dz.$		3				
829	06	Evaluate: $\int_0^1 \int_0^{\sqrt{z}} \int_0^{2\pi} (r^2 \cos^2 \theta + z^2) r d\theta dr dz.$		3				
830	06	Evaluate: $\int_0^1 \int_0^{2-x} \int_0^{2-x-y} dz dy dx.$		2				
831	06	Evaluate: $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} z dz dy dx$		4				
832	06	Evaluate: $\int_1^3 \int_{1/x}^1 \int_0^{\sqrt{xy}} xyz dz dy dx.$		2				
833	06	Evaluate the integral $\int_0^2 \int_0^2 \int_0^{yz} xyz dx dy dz.$		4				
834	06	Evaluate the triple integral. $\int_0^1 \int_0^\pi \int_0^\pi y \sin z dx dy dz$		3				
835	06	Evaluate: $\int_0^1 \int_0^{1-x} \int_1^{(x+y)^2} x dz dy dx.$		4				
836	06	Evaluate $\int \int \int z^2 dx dy dz$ over the region common to the sphere $x^2 + y^2 + z^2 = 4$ and the cylinder $x^2 + y^2 = 2x$		7				
837	06	Evaluate: $\iiint (x - 2y + z) dx dy dz$ over the region R where $R: 0 \leq x \leq 1, 0 \leq y \leq x^2, 0 \leq z \leq x + y$		2				
838	06	Evaluate: $\int_1^e \int_1^{\log y} \int_1^{e^x} \log z dz dx dy$		4				
839	06	Evaluate: $\int_0^1 \int_0^y \int_0^{x+2y} (x + y + z) dz dx dy.$		3				
840	06	Evaluate: $\int_0^a \int_0^x \int_0^{x+y} e^{x+y+z} dz dy dx.$		4				
841	06	Evaluate: $\int_0^1 \int_0^y \int_0^{x+y} y dz dx dy.$		3				
842	06	Evaluate: $\int_{x=0}^1 \int_{y=0}^x \int_{z=0}^{\sqrt{x+y}} z dz dy dx.$		3				
843	06	Evaluate $\int_{-1}^1 \int_0^2 \int_0^1 (xz - y^3) dz dy dx.$		3				
844	06	Evaluate $\int \int \int xyz dx dy dz$ over the positive octant of the sphere $x^2 + y^2 + z^2 = 4$		7				
845	06	Evaluate: $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} \frac{1}{(x+y+z+1)^3} dz dy dx$		3				
846	06	Evaluate $\int_0^1 \int_0^{1-x} \int_0^{x+y} dz dy dx$		4				
847	06	Evaluate $\int_0^2 \int_0^y \int_{x-y}^{x+y} (x + y + z) dz dx dy$		3				
848	06	Find the area of region R enclosed by the line $y = x, x = 1$ and the $x - axis$.		4				
849	06	Find the area of the circle $x^2 + y^2 = 1$ using double integration		3				
850	06	Find the area between parabolas $x^2 = 4ay$ and $y^2 = 4ax$ where a is constant.		5				

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851	06	Find the area of the region R enclosed by the parabola the line $y = x^2$ and $y = x + 2$.		5				
852	06	Evaluate $\iint (x + y)^2 dx dy$ over the curve bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$		5				
853	06	Using double integration find area of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$		4				
854	06	Find the area of region enclosed by $y = x^2, y = 2x - x^2$		4				
855	06	Evaluate $\iint_D (xy) dA$, where D is the region bounded by the line $y = x - 1$ and parabola $y^2 = 2x + 6$		4				
856	06	Evaluate $\int \int \frac{2xy^5}{\sqrt{1+x^2y^2-y^4}} dx dy$ over the triangle with vertices (0,0), (1,1), (0,1)		4				
857	06	Find the area between the rectangular hyperbola $3xy = 2$ and the line $12x + y = 6$.		4				
858	06	Find the area common to $r = a$ and $r = 2a \cos \theta; a > 0$		3				
859	06	Find the area common to the cardioid $r = a(1 + \cos \theta), r = a(1 - \cos \theta)$.		4				
860	06	Using double integrations, find the area enclosed by the cardioid $r = a(1 + \cos \theta)$.		4				
861	06	Find the area lying inside the cardioid $r = a(1 + \cos \theta)$.and outside the region of the circle $r = a$, by double integration.		4				
862	06	Find the area outside the circle $r = 2a \cos \theta$ and inside the cardioid $r = a(1 + \cos \theta)$.		3				
863	06	Evaluate the area between the circles $r = 2 \cos \theta$ and $r = 4 \cos \theta$, using double integration.		4				
864	06	Find the area common to both of the circles $r = \sin \theta$ and $r = \cos \theta$		4				
865	06	Find the area lying inside the circle $r = a \sin \theta$ and outside the cardioids $r = a(1 - \cos \theta)$		4				
866	06	Evaluate $\iint r^3 dr d\theta$ over the area bounded between the circles $r = 4 \cos \theta$ and $r = 9 \cos \theta$.		3				
867	06	Change into spherical polar coordinates and hence evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} \frac{dz dy dx}{\sqrt{a^2-x^2-y^2-z^2}}$		4				
868	06	Evaluate $\iiint_D z \sqrt{x^2 + y^2} dV$ where D is the solid bounded by the surfaces $x^2 + y^2 = 3z, z = 3$.		5				
869	07	If $\vec{a} = 2\vec{i} - 3\vec{j} + \vec{k}$ then the $ \vec{a} =$ _____	$\sqrt{14}$	1	$\sqrt{-4}$	$\sqrt{4}$	$\sqrt{13}$	$\sqrt{14}$
870	07	$\vec{i} \times \vec{j}$ is _____	$\vec{k}$	1	$\vec{k}$	$-\vec{k}$	0	none of these
871	07	If $\vec{u} = 6\vec{i} - 3\vec{j} + 2\vec{k}$ then $ u $ is	$\sqrt{49}$	1	$\sqrt{49}$	$-\sqrt{49}$	49	none of these
872	07	If $\vec{a} \cdot \vec{b} = 0$ then angle between $\vec{a}$ and $\vec{b}$ is	none of these	1	0	2π	π	none of these
873	07	If $\vec{a} = 2\vec{i} + 3\vec{j} - \vec{k}, \vec{b} = -\vec{i} + 2\vec{j} - 4\vec{k}$ and $\vec{c} = \vec{i} + \vec{j} + \vec{k}$, then $(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c}) =$ _____	-74	1	16	8	74	-74
874	07	For which value of k are $u = (k, k, 1)$ and $v = (k, 5, 6)$ is orthogonal?		2				
875	07	If $r = t^3 i + \left(2t^3 - \frac{1}{5t^2}\right) j$ then show that $r \times \frac{dr}{dt} = \hat{k}$		3				
876	07	Give parametric representations of each of the following curves: -		3				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
		1) The line $9x - 5y = 7$ 2) The parabola $y = (x - 2)^2$ The circle $x^2 + y^2 = 4$						
877	07	Find the arc length of the curve $\vec{r}(t) = (1 + 3t^2)\hat{i} + (4 + 4t^3)\hat{j}$ from $t=0$ to $t=1$.		3				
878	07	Find the length of arc of the curve $\vec{r}(t) = \frac{2\sqrt{2}}{3}t^3\hat{i} + \frac{t^2}{2}\hat{j} + (t + 3)\hat{k}$ between $t = 0$ and $t = 2$		2				
879	07	Find the length of the arc of the curve $y = \log \sec x$ from $x = 0$ to $x = \frac{\pi}{3}$.		4				
880	07	Find the arc length of the portion of the circular helix $\vec{r}(t) = \cos t\hat{i} + \sin t\hat{j} + t\hat{k}$ from $t=0$ to $t=\pi$.		4				
881	07	The shape of a cable from an antenna tower is given by the equation $y = \frac{4}{3}x^{\frac{3}{2}}$ from $x = 0$ to $x = 20$. Find the total length of the cable.		4				
882	07	The value of $ \nabla\phi $ if $\phi = 2xz^4 - x^2y$ at $(2,-2,-1)$ is	$2\sqrt{93}$	1	$5\sqrt{93}$	$5\sqrt{85}$	$5\sqrt{99}$	$2\sqrt{93}$
883	07	If $\phi = x + y + z$, then the value of $ \text{grad}\phi $ at $(1, 2, -1)$ is (a) 0 (b) 1 (c) 2 (d) 3(e) $\sqrt{2}$ (f)	$\sqrt{3}$	1	$\sqrt{3}$	3	$\sqrt{2}$	$\sqrt{6}$
884	07	If $\phi = 3x^2y - y^3z^2$ then $\text{grad}\phi$ at point $(1, -2, 1)$ is _____	$-12\hat{i} - 9\hat{j} + 16\hat{k}$	1	$-12\hat{i} - 9\hat{j} - 16\hat{k}$	$12\hat{i} + 9\hat{j} + 16\hat{k}$	$2\hat{i} + 9\hat{j} + 6\hat{k}$	None of These
885	07	If $\phi = x^2 + y^2 + z^2$, then the value of $ \text{grad}\phi $ at $(1, 2, -2)$ is	6	1	2	4	6	8
886	07	If $\phi = x^2 + y^2 + z^2$, then the value of $ \text{grad}\phi $ at $(1, -1, \frac{1}{2})$ is	3	1	0	3	1	$\sqrt{2}$
887	07	If $\phi = xyz$, then the value of $ \text{grad}\phi $ at $(1, 2, -1)$ is	3	1	0	1	2	3
888	07	If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, Prove that $\nabla^2 r^n = n(n+1)r^{n-2}$		3				
889	07	Prove that $\nabla^2 f(r) = f''(r) + \frac{2}{r}f'(r)$		3				
890	07	If $\phi = 3x^2y - y^3z^2$, find $\text{grad}\phi$ at the point $(1, -2, -1)$.		3				
891	07	Find a unit vector normal to the surface $x^3 + y^3 + 3xyz = 3$ at the point $(1, 2, -1)$		4				
892	07	If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, show that $\nabla \log r = \frac{1}{r}\hat{r}$. Where $\hat{r}$ is unit vector.		3				
893	07	Find the gradient of $f(x, y, z) = 2z^3 - 3(x^2 + y^2)z + \tan^{-1}(xz)$ at $(1, 1, 1)$.		4				
894	07	Find values of constants λ and μ so that the surfaces $\lambda x^2 - \mu yz = (\lambda + 2)x$ and $4x^2y + z^3$ intersect orthogonally at the point $(1, -1, 2)$.		3				
895	07	Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $r = \vec{r} $ and $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ is a constant vector. Prove that $\nabla \left(\frac{\vec{a} \cdot \vec{r}}{r^n} \right) = \frac{\vec{a}}{r^n} - \frac{n(\vec{a} \cdot \vec{r})}{r^{n+2}} \vec{r}$		2				
896	07	Find the angle between the normal to the surfaces $xy = z^2$ at $P(1, 1, 1)$ and $Q(4, 1, 2)$.		3				
897	07	Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$.		3				
898	07	The directional derivative of $f(x, y) = xy^2 - y^3$ at $P(2, 2)$ in the direction of $\theta = \frac{\pi}{4}$ is	0	1	1	0	$4\sqrt{2}$	4

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899	07	The magnitude of the maximum directional derivative of the function $2x + y + 2z$ at the point (1,0,0) is	3	1	0	1	2	3
900	07	Define gradient of a function. Use it to find directional derivative of $f(x, y, z) = x^3 - xy^2 - z$ at $P(1,1,0)$ in the direction of $\vec{a} = 2\hat{i} - 3\hat{j} + 6\hat{k}$		3				
901	07	Find the directional derivative of $\phi = xy^2 + yz^3$ at $(2, -1, 1)$ in the direction of the normal to the surface $\log z - y^2 = -4$ at $(-1, 2, 1)$.		4				
902	07	Find the directional derivative of $\phi = \frac{1}{(x^2 + y^2 + z^2)^{\frac{1}{2}}}$ at point $P(1, -1, 1)$ in the direction of $\vec{a} = \hat{i} + \hat{j} + \hat{k}$		3				
903	07	Find the directional derivative of $f(x, y, z) = \cos xy + e^{yz} + \ln xz$ at a point $P(1, 0, 1/2)$ in the direction $\vec{u} = \hat{i} + 2\hat{j} + 2\hat{k}$		4				
904	07	Find the directional derivative of the divergence of $\vec{F}(x, y, z) = xy^2\hat{i} + zy^2\hat{j} + xz^2\hat{k}$ at the point (2, 0, 3) in the direction of the outer normal to the sphere $x^2 + y^2 + z^2 = 14$ at the point (3, 2, 1).		4				
905	07	Find the direction in which the directional derivative of $\phi = \frac{x^2 - y^2}{xy}$ at (1, 1) is zero.		3				
906	07	Find the directional derivative of $f(x, y, z) = 2x^2 + 3y^2 + z^2$ at point $P(2, 1, 3)$ in the direction of $\vec{a} = [1, 0, -2]$.		2				
907	07	Find the derivative of $f(x, y) = x^2 \sin 2y$ at the point $(1, \frac{\pi}{2})$ in the direction of $\vec{v} = 3\hat{i} - 4\hat{j}$.		3				
908	07	Find the directional derivative of $x^2y^2z^2$ at (1, 1, -1) along a direction equally inclined with coordinate axes.		4				
909	07	Find the directional derivative of $\phi = 4xz^3 - 3x^2yz^2$ at the point (2, -1, 2) in the direction $2\hat{i} + 3\hat{j} + 6\hat{k}$		4				
910	07	Find the derivative of $f(x, y) = xe^y + \cos xy$ at the point (2, 0) in the direction of $A = 3\hat{i} - 4\hat{j}$.		2				
911	07	Find the directional derivative of $\phi = x^2 - y^2 + 2z^2$ at the point P (1, 2, 3) in the direction of the line PQ where Q is the point (5, 0, 4).		4				
912	07	Find the directional derivative of $f(x, y, z) = x^3 - xy^2 - z$ at (1, 1, 0) in the direction of $2\hat{i} - 3\hat{j} + 6\hat{k}$		4				
913	07	Find the directional derivative of the function $f(x, y, z) = xy^2 + yz^3$ at the point (2, -1, 1) in the direction of the vector $\hat{i} + 2\hat{j} + 2\hat{k}$.		4				
914	07	Find directional derivative of the function $f(x, y) = ax + by$; a, b are constants, at the point P(0, 0) which makes an angle of 30° with positive x-axis.		3				
915	07	Find the directional derivative of the divergence of $\vec{F}(x, y, z) = xy\hat{i} + xy^2\hat{j} + z^2\hat{k}$ at the point (2, 1, 2) in the direction of		3				

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		the outer normal to the sphere $x^2 + y^2 + z^2 = 9$.						
916	07	Find the directional derivative of the scalar point function $f = 3e^{2x-y+z}$ at the point $A(1,1,-1)$ in the direction towards the point $B(-3,5,6)$		3				
917	07	Find the directional derivative $D_u f(x,y)$ if $f(x,y) = x^3 - 3xy + 4y^2$ and u is the unit vector given by angle $\theta = \frac{\pi}{6}$. What is $D_u f(1,2)$?		3				
918	07	The temperature at any point in space is given by $T = xy + yz + zx$. Determine the derivative of T in the direction of the vector $3\hat{i} - 4\hat{k}$ at the point $(1,1,1)$.		4				
919	07	Find directional derivative of the function $\phi = \frac{x}{x^2+y^2}$ at the point $P(0,2)$ which makes an angle of 30° with positive x-axis.		3				
920	07	If $r = xi + yj + zk$ then $\text{div}(r)$ is	3	1	r	0	1	3
921	07	The divergence of $\vec{F} = xyz\hat{i} + 3x^2y\hat{j} + (xz^2 - y^2z)\hat{k}$ is	$yz + 3x^2 + (2xz - y^2)$	1	$yz + 3x + 2xz$	$yz + xy$	$yz + 3x^2 + (2xz - y^2)$	$xy - yz$
922	07	If $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$ then $\nabla \cdot \vec{F}$ at $(1,1,1)$ is	3	1	0	-1	3	none of these
923	07	Find the value of 'a' if $F = (x + 3y)i + (y - 2z)j + (az + x)k$ is solenoidal	-2	1	2	1	-2	0
924	07	For vector point function $\vec{F}$, divergence of $\vec{F}$ is obtained by	$\nabla \cdot \vec{F}$	1	$\nabla \cdot \vec{F}$	$\nabla \times \vec{F}$	$\nabla \vec{F}$	$\nabla^2 \vec{F}$
925	07	If $\text{div } \vec{u} = 0$ then $\vec{u}$ is said to be	Solenoidal	1	Rotational	Solenoidal	Compressible	None of these
926	07	If $\phi = \tan^{-1}\left(\frac{y}{x}\right)$ then $\text{div}(\text{grad } \phi)$ is	0	1	0	1	2	4
927	07	If $\vec{F} = (x + 3y^2)\hat{i} + (2y + 2z^2)\hat{j} + (x^2 + az)\hat{k}$ is solenoidal then the constant a is _____.	-3	1	0	2	-3	-2
928	07	If $\vec{F} = (bx + 4y^2z)\hat{i} + (x^3 \sin z - 3y)\hat{j} - (e^x + 4 \cos x^2y)\hat{k}$ is solenoidal, then $b =$ _____	3	1	4	0	2	3
929	07	The value of 'a' is _____, if $F = (2x + 3y)i + (3y + 2z)j + (az + x)k$ is solenoidal	-5	1	0	-1	-5	3
930	07	If $\vec{a} = 2x\hat{i} + 3y\hat{j} + 4z\hat{k}$ and $u = x^2 + y^2 + z^2$, then $\text{div}(u\vec{a})$ at $(1,1,1)$ is _____.	45	1	45	55	-45	-55
931	07	$\vec{F}$ is solenoidal vector, If $\text{div}(\vec{F})$ is _____	0	1	1	0	-1	2
932	07	If $\phi = xyz - 2y^2z + x^2z^2$, Find $\text{div}(\text{grad } \phi)$ at the point $(2,4,1)$.		3				
933	07	Find the value of the constant λ such that the vector field defined by $F = (2x^2y^2 + z^2)i + (3xy^3 - x^2z)j + (\lambda xy^2z + xy)k$ is solenoidal.		3				

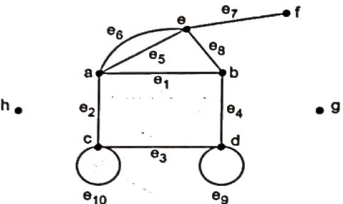
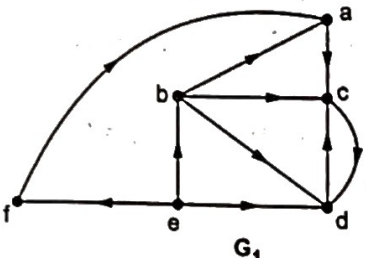
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934	07	Determine whether the vector field $\vec{u} = y^2\hat{i} + 2xy\hat{j} - z^2\hat{k}$ is solenoidal at a point (1,2,1).		3				
935	07	Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $r = \vec{r} $ and $\vec{a}$ is a constant vector. Find the value of $\text{div}\left(\frac{\vec{a}\times\vec{r}}{r^n}\right)$		5				
936	07	If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, Show that $\text{div}(\text{grad } r^n) = n(n+1)r^{n-2}$.		3				
937	07	If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, Show that $\text{Div}(r^n\vec{r}) = (n+3)r^n$.		3				
938	07	If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $r = \vec{r} $ and $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ are constant vectors. Prove that $\vec{a} \cdot \nabla \left(\vec{b} \cdot \nabla \frac{1}{r} \right) = \frac{3(\vec{a} \cdot \vec{r})(\vec{b} \cdot \vec{r})}{r^5} - \frac{\vec{a} \cdot \vec{b}}{r^3}$		4				
939	07	Find divergence and curl of $\vec{v} = xyz[x, y, z]$.		2				
940	07	The velocity vector $\vec{v} = \vec{r}'(t) = x^3\hat{k}$ of a fluid motion is given. Is the flow irrotational? Incompressible?		3				
941	07	Find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$, where $\vec{F} = \text{grad}(x^3 + y^3 + z^3 - 3xy)$		7				
942	07	If $\vec{F}$ is conservative field then $\text{curl } \vec{F} =$ _____	$\vec{0}$	1	i	j	k	$\vec{0}$
943	07	If $\vec{F}$ is irrotational then	$\nabla \times \vec{F} = \vec{0}$	1	$\nabla \vec{F} \neq \vec{0}$	$\nabla \times \vec{F} = \vec{0}$	$\nabla \cdot \vec{F} = 0$	none of these
944	07	The value of $\text{curl}(\text{grad } \phi)$, where $\phi = 2x^2 - 3y^2 + 4z^2$ is	0	1	4xi -6yj +8zk	4x - 6y + 8z	0	6
945	07	If $f(x, y, z) = xyz$ then $\text{Curl}(\text{grad } f) =$ _____	0	1	0	X	$xi + yj + zk$	xyz
946	07	If $\vec{F}(x, y, z) = x\hat{i} + y\hat{j} + z\hat{k}$ then $\text{curl } \vec{F}$ is _____	0	1	1	3	2	0
947	07	If curl of $(2xy+3yz)\hat{i} + (x^2+axz-4z^2)\hat{j} + (3xy+2byz)\hat{k}$ is zero then value of a and b is	a = 3 & b = -4	1	a = 3 & b = -4	a = 3 & b = 4	a = 9 & b = 13	a = -3 & b = -4
948	07	The value of $\text{curl}(\text{grad } \phi)$, where $\phi = 2x^2 - 3y^2 + 4z^2$ is _____	0	1	$4x\hat{i} - 6y\hat{j} + 8z\hat{k}$	0	3	$4x\hat{i} + 6y\hat{j} + 8z\hat{k}$
949	07	What is the value of $\text{curl}(\vec{a} \times \vec{r})$? Where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$	$2\vec{a}$	1	$-\vec{r}$	$\vec{a}$	$2\vec{a}$	$-\vec{a}$
950	07	Show that $\vec{F} = \frac{(y^2 - x^2)\hat{j}}{x^2 + y^2}$ is irrotational.		3				
951	07	Find $\text{curl } \vec{F}$ at the point (2,0,3), if $\vec{F} = z e^{2xy}\hat{i} + 2xy \cos y \hat{j} + (x + 2y)\hat{k}$.		3				
952	07	Find $\text{curl } F$, if $F = (y^2 \cos x + z^3)\hat{i} + (2y \sin x - 4)\hat{j} + 3xz^2\hat{k}$. Whether F is irrotational?		3				
953	07	Prove that $\vec{F} = (z^2 + 2x + 3y)\hat{i} + (3x + 2y + z)\hat{j} + (y + 2zx)\hat{k}$ is irrotational. Also find scalar potential ϕ such that $\vec{F} = \text{grad } \phi$ and $\phi(1,1,0) = 4$		3				
954	07	Show that $\vec{F} = (y^2 - z^2 + 3yz - 2x)\hat{i} + (3xz + 2xy)\hat{j} + (3xy - 2xz + 2z)\hat{k}$ is both solenoidal and irrotational		3				
955	07	Find constants a,b,c, so that $V = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy +$		3				

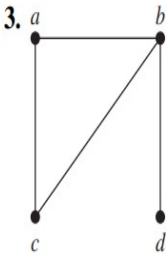
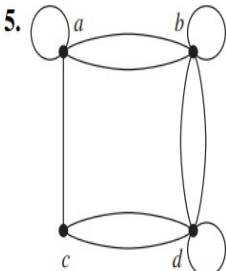
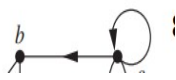
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		2z) is irrotational. Also, find scalar potential function.						
956	07	Find the divergence and curl of $\vec{V} = e^{xyz}(\hat{i} + \hat{j} + \hat{k})$ at the point (1,2,3).		4				
957	07	Calculate the curl of the vector $F = xyz \hat{i} + 3x^2y \hat{j} + (xz^2 - y^2z) \hat{k}$		3				
958	07	Prove that $\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$, for any vector $\vec{A} = A_1\hat{i} + A_2\hat{j} + A_3\hat{k}$.		5				
959	07	A vector field is given by $\vec{F} = (x^2 + xy^2)\vec{i} + (y^2 + x^2y)\vec{j}$. Find the scalar potential. Show that $\vec{F}$ is irrotational.		7				
960	07	Show that the vector field $F = (y \sin z - \sin x)i + (x \sin z + 2yz)j + (xycos z + y^2)k$ is conservative and find the corresponding scalar potential.		5				
961	07	Show that $\vec{F} = 2xyz\vec{i} + (x^2z + 2y)\vec{j} + x^2y\vec{k}$ is conservative. Find its scalar potential function ϕ .		4				
962	07	Show that $\vec{F} = (e^x \cos y + yz)i + (xz - e^x \sin y)j + (xy + z)k$ is conservative and find the potential function.		4				
963	07	Show that the $\vec{F} = (6xy + z^3)i + (3x^2 - z)j + (3xz^2 - y)k$ is irrotational. Find the scalar ϕ such that $\vec{F} = \nabla \phi$.		4				
964	07	Prove $\vec{F} = (y^2 \cos x + z^3)\hat{i} + (2y \sin x - 4)\hat{j} + 3xz^2\hat{k}$ that is irrotational and find its scalar potential.		5				
965	07	Check whether the vector field $\vec{F} = e^{y+2z}\hat{i} + x e^{y+2z}\hat{j} + 2x e^{y+2z}\hat{k}$ is conservative or not. If yes, find the scalar potential function $\phi(x,y,z)$ such that $\vec{F} = \text{grad } \phi$.		7				
966	07	Show that $\vec{F} = (x^2 - yz)\hat{i} + (y^2 - zx)\hat{j} + (z^2 - xy)\hat{k}$ is conservative. Find its scalar potential ϕ .		5				
967	08	If the value of the line integral $\oint_C \vec{F} \cdot d\vec{r}$ does not depend on path C then $\vec{F}$ is	irrotational	1	solenoidal	incompressible	irrotational	none of these
968	08	If $\vec{F} = 3xy\hat{i} - y^2\hat{j}$; evaluate $\int_C \vec{F} \cdot d\vec{r}$ Where c is the arc of the Parabola $y = 2x^2$ from (0, 0) to (1, 2).	-7/6	1	-7/6	2/3	-2/3	7/6
969	08	Line integral along straight line from (1,0) to (1,2) for $\vec{F} = 2\hat{i} + \hat{j}$ is _____.	2	1	2	1	-2	-3
970	08	Evaluate $\int_C ydx + xdy + zdz$, where C is given by $x = \cos t, y = \sin t, z = t^2, 0 \leq t \leq 2\pi$.		4				
971	08	If $\vec{F} = (2xy + z^3)i + x^2j + 3xz^2k$. Show that $\oint_C \vec{F} \cdot d\vec{r}$ is independent of path of integration. Hence find the integral when C is any path joining (1, -2, 1) and (3, 1, 4)		4				
972	08	Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = (x^2 + y^2)i + (x^2 - y^2)j + z^2k$ along the boundary of the triangle with vertices (1,1,0), (0,1,0), (0,0,0).		4				

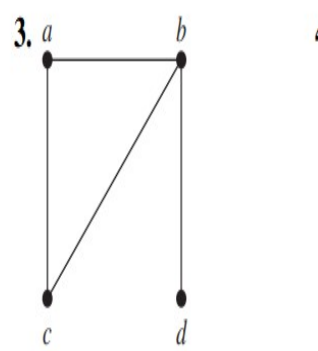
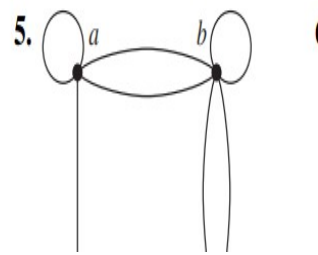
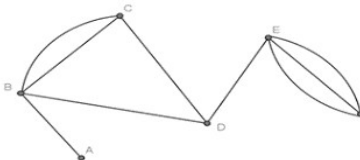
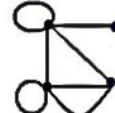
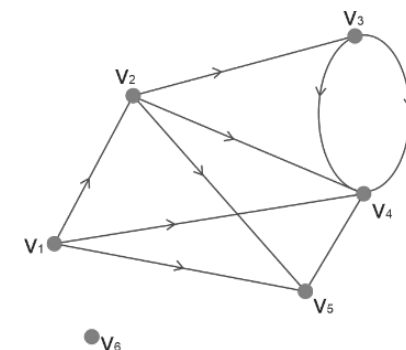
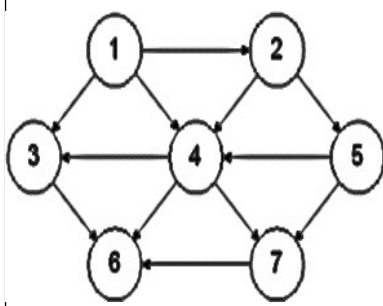
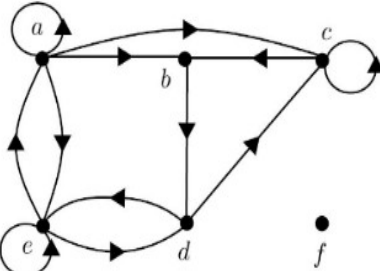
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973	08	Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = (x^2 - y^2)\hat{i} + 2xy\hat{j}$ and C is the curve given by the parametric equation $C:r(t) = t^2\hat{i} + t\hat{j}; 0 \leq t \leq 2$.		3				
974	08	Integrate $f(x,y,z) = x - yz^2$ over the curve $C = C_1 + C_2$, where C_1 is the line segment joining (0,0,1) to (1,1,0) and C_2 is the curve $y = x^2$ joining (1,1,0) to (2,4,0).		7				
975	08	Evaluate $\int_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ Where, C is the rectangle in XY-plane bounded by $y = 0, x = a, y = b, x = 0$		7				
976	08	$\int_{(1,2)}^{(3,4)} (xy^2 + y^3)dx + (x^2y + 3xy^2)dy$ is independent of path joining the points (1, 2) and (3,4). Hence, evaluate the integral.		7				
977	08	Evaluate $\int_C \vec{F} \cdot d\vec{r}$ along the parabola $y^2 = x$ between the points (0,0) and (1,1), where $\vec{F} = x^2\hat{i} + xy\hat{j}$.		3				
978	08	Evaluate $\int_C \vec{F} \cdot d\vec{r}$ along the curve $x^2 + y^2 = 1, z=l$ in the positive direction from (0,1,1) to (1,0,1) where $\vec{F} = (yz + 2x)\hat{i} + xz\hat{j} + (xy + 2z)\hat{k}$		3				
979	08	Evaluate $\int_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = (x^2 + xy)\hat{i} + (x^2 + y^2)\hat{j}$ Where, C is the rectangle in XY-plane bounded by $y = \pm 1, x = \pm 1$.		3				
980	08	If $\vec{F} = (2xy + z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$. Show that $\int_C \vec{F} \cdot d\vec{r}$ is independent of path of integration. Hence find the integral when C is any path joining (1, -2,1) and (4,2,5)		3				
981	08	If $\vec{F} = (y^2 + 2xz)\hat{i} + (z^2 + 2xy)\hat{j} + (x^2 + 2yz)\hat{k}$. Show that $\oint_C \vec{F} \cdot d\vec{r}$ is independent of path of integration. Hence find the integral when C is any path joining (1,0,1) and (1,2,3).		3				
982	08	If $\vec{F} = 5xy\hat{i} + 2y\hat{j}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is the part of the curve $y = x^3$ between $x = 1$ and $x = 2$.		3				
983	08	If $\vec{F} = (2xy + z^2)\hat{i} + x^2\hat{j} + 2xz\hat{k}$. Show that $\oint_C \vec{F} \cdot d\vec{r}$ is independent of path of integration. Hence find the integral when C is any path joining (1, -2,1) and (3,1,4)		3				
984	08	Show that $\int_{(1,1,1)}^{(2,2,2)} \vec{F} \cdot d\vec{r}$ is independent of path for $\vec{F} = (x^2 - yz)\hat{i} + (y^2 - zx)\hat{j} + (z^2 - xy)\hat{k}$. Hence evaluate the integral.		3				
985	08	If $\vec{F} = \frac{y^2 - x^2}{x^2 + y^2}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$, where C is the circle $x^2 + y^2 = 1$ traversed counter-clockwise.		3				
986	08	If $\vec{F} = (x^2 - yz)\hat{i} + (y^2 - zx)\hat{j} + (z^2 - xy)\hat{k}$, then (i) prove that $\vec{F}$ is conservative (ii) find its scalar potential ϕ find $\oint_C \vec{F} \cdot d\vec{r}$, along any path joining (1,1,0) and (2,0,1)		3				
987	08	Evaluate $\int_C (2x^2 - y^2)dx + (x^2 + y^2)dy$ where C is the boundary of the area enclosed by the x-axis and the upper half of the circle $x^2 + y^2 = a^2$.		4				
988	08	Find the work done when a force $\vec{F} = (x^2 - y^2 + x)\hat{i} - (2xy + y)\hat{j}$ moves a particle		4				

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		in the XY plane from (0,0) to (1,1) along the parabola $y^2 = x$.						
989	08	Find the work done by $\vec{F} = (y - x^2) \hat{i} + (z - y^2) \hat{j} + (x - z^2) \hat{k}$ over the curve $\vec{r}(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$; $0 \leq t \leq 1$, from (0,0,0) to (1,1,1).		3				
990	08	Find the work done when a force $\vec{F} = (2x^2y\hat{i} + 3xy\hat{j})$ moves a particle in the XY plane from (0,0) to (1,4) along the parabola $y = 4x^2$		2				
991	08	Find the work done when a force $\vec{F} = (x^2 - y^2 + 2x)\hat{i} - (2xy + y)\hat{j}$ moves a particle in the xy plane from (0,0) to (1,1) along the parabola $y^2 = x$. Is the workdone different when the path is straight line $y = x$?		4				
992	08	Find the work done by the force $\vec{F} = (3x^2 - 3x)\hat{i} + 3z\hat{j} + k\hat{k}$ along the straight line $\vec{r}(t) = t\hat{i} + t\hat{j} + t\hat{k}$, $0 \leq t \leq 1$.		3				
993	08	Find the work done in moving a particle in the force field $\vec{F} = 3x^2\hat{i} + (2xz - y)\hat{j} + z\hat{k}$ along the straight line from (0,0,0) to (2,1,3).		3				
994	08	If $\int_C \vec{F} \cdot d\vec{r} = \int_C e^{x-y+z^2} (dx - dy + 2zdz)$. Show that $\oint_C \vec{F} \cdot d\vec{r}$ is independent of path of integration. Hence find its scalar potential ϕ and find the work done in moving a particle under this force field from A(0,-1,1) to B(2,4,0).		3				
995	08	Find the work done by the force $\vec{F} = x\hat{i} - z\hat{j} + 2y\hat{k}$ in displacing the particle along the triangle OAB, where $OA: 0 \leq x \leq 1, y = x, z = 0$ $AB: 0 \leq z \leq 1, x = 1, y = 1$ $BO: 0 \leq x \leq 1, y = z = x$		3				
996	08	Find the work done by the force $\vec{F} = x^2y\hat{i} + \frac{x^3}{3}\hat{j} + xy\hat{k}$ along the curve $\cos t\hat{i} + \sin t\hat{j} + (2\sin^2 t - 1)\hat{k}$, $0 \leq t \leq 2\pi$.		3				
997	08	Using line integral, evaluate the work done by the force $\vec{F} = (2y + 3)\hat{i} + xz\hat{j} + (yz - x)\hat{k}$ when it moves a particle from point (0, 0, 0) to (2, 1, 1) along the curve $x = 2t^2, y = t, z = t^3$.		3				
998	08	Find the work done in moving a particle in the force field $\vec{F} = 3x^2\hat{i} + (2xz - y)\hat{j} + z\hat{k}$ along the curve $x^2 = 4y$ and $3x^3 = 8z$ from $x = 0$ to $x = 2$		3				
999	08	Find constants a, b, c, so that $\vec{v} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$ is irrotational. Also, find scalar potential function and the work done in moving a particle under this force field from (1, 1, 0) to (2, 0, 1)		3				
1000	08	Find the work done by the force $\vec{F} = 16y\hat{i} + (3x^2 + 2)\hat{j}$ in moving a particle once round the right half of the ellipse $x^2 + y^2/a^2 = a^2$ from (0,1) to (0,-1).		3				
1001	08	Find the work done in moving a particle along the straight-line segments joining the points (0,0,0) to (1,0,0) then to (1,1,0) and finally to (1,1,1) under the force field $\vec{F} = (3x^2 + 6y)\hat{i} - 14yz\hat{j} + 20xz^2\hat{k}$		3				
1002	08	If $\vec{F} = 4\hat{i} + 0\hat{j}$, the flux of $\vec{F}$ through the line segment from (4,0) to (0,4) = _____	16	1	16	8	4	-8

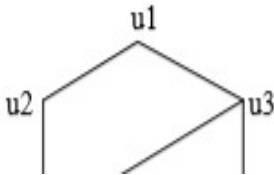
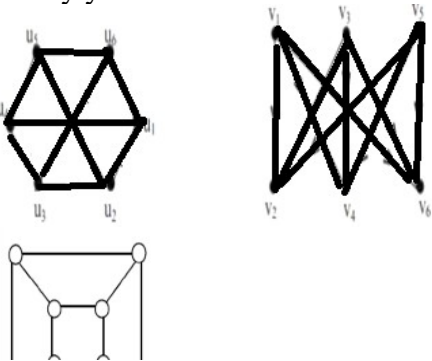
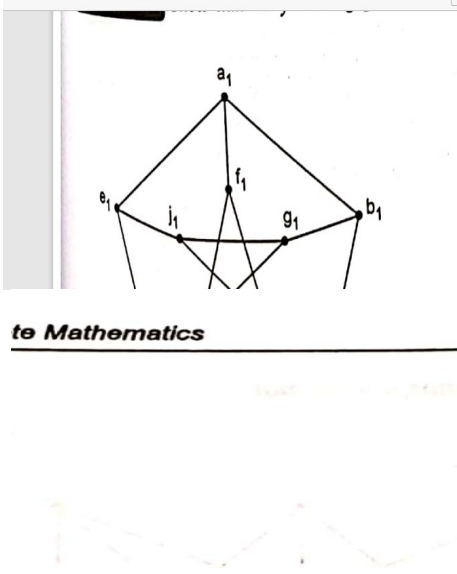
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1003	08	If $\vec{F} = (2x - y + 2z)\vec{i} + (x + y - z)\vec{j} + (3x - 2y - 5z)\vec{k}$ calculate circulation of $\vec{F}$ along the circle in the xy plane of 2 unit radius and center at the origin.		7				
1004	08	If $\vec{F} = 2\vec{i} + 0\vec{j}$ calculate the flux of $\vec{F}$ through the line segment from (3,0) to (0,3)		4				
1005	08	Find the flux of $\vec{F} = 3xy\vec{i} + (x - y)\vec{j}$ through the parabolic arc $y = x^2$ between (-1,1) and (4,16)		4				
1006	08	Find the flux of $\vec{F} = 3x\vec{i} + 5y\vec{j}$ through the circle $x^2 + y^2 = 1$		4				
1007	08	Find flux of $\vec{F} = x^2\vec{i} - y\vec{j}$ through circle of radius 3 units and centre at origin.		4				
1008	08	Verify Green's theorem for vector function $\vec{F} = (y^2 - 7y)\vec{i} + (2xy + 2x)\vec{j}$ and curve C: $x^2 + y^2 = 1$.		7				
1009	08	Verify Green's theorem for $\oint (3x^2 - 8y^2)dx + (4y - 6xy)dy$ and the path is boundary of the triangle with vertices (0,0), (1,0) and (0,1).		7				
1010	08	State Green's theorem and also evaluate the integral $\oint (6y + x)dx + (y + 2x)dy$ where C: the circle $(x - 2)^2 + (y - 3)^2 = 4$.		5				
1011	08	Use Green's theorem to evaluate $\int_C [x^2y dx + y^3 dy]$, where C is the closed path formed by $y = x$ and $y = x^3$ from (0,0) to (1,1).		5				
1012	08	Using Green's theorem evaluate $\oint_C xydy - y^2dx$ where C is the square cut from the first quadrant by the lines $x = 1$ and $y = 1$.		5				
1013	08	Verify Green's theorem for $\vec{F} = x^2\vec{i} + xy\vec{j}$ under the square bounded by $x = 0, x = 1, y = 0, y = 1$.		7				
1014	08	Verify Green's theorem for the function $\vec{F} = (x + y)\vec{i} + 2xy\vec{j}$ and C is the rectangle in the xy -plane bounded by $x = 0, y = 0, x = a, y = b$.		7				
1015	08	Using Green's theorem evaluate $\oint (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the boundary of the region bounded by $y^2 = x$ and $x^2 = y$		7				
1016	08	State the Green's theorem and use it to evaluate the integral $\oint y^2dx + x^2dy$, where c is the triangle bounded by $x = 0, x + y = 1, y = 0$.		7				
1017	08	Verify Green's theorem for $\oint_C [(y - \sin x)dx + \cos x dy]$: where C is the plane triangle enclosed by the lines $y = 0, x = \frac{\pi}{2}, y = \frac{2x}{\pi}$		7				
1018	08	Verify Green's theorem for the function $\vec{F} = (x^2 + y^2)\vec{i} - 2xy\vec{j}$, where C is the rectangle in the xy -plane bounded by $y = 0, y = b, x = 0$ and $x = a$.		7				

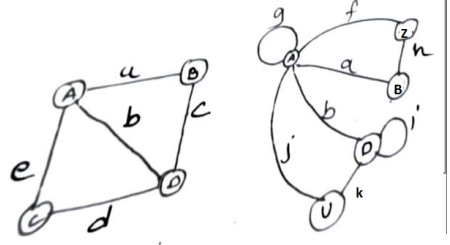
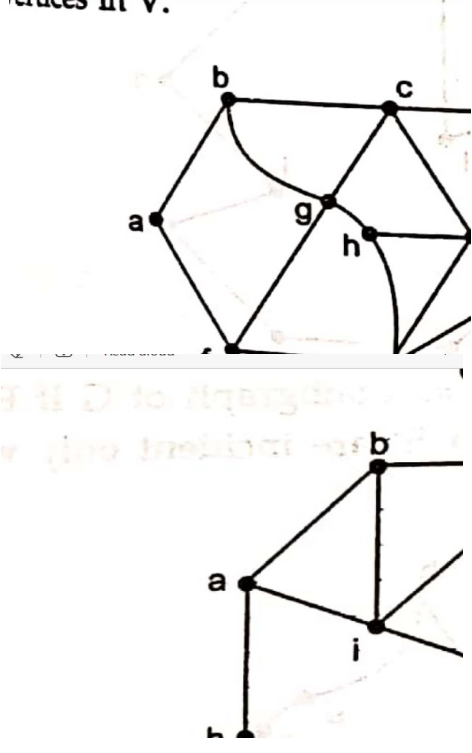
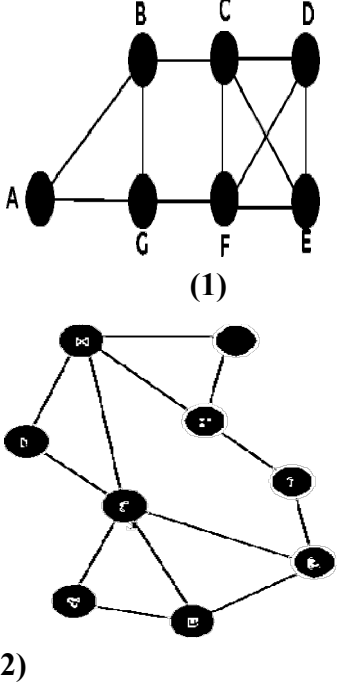
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1019	08	Verify Green's theorem for the field $\vec{F} = (x - y)\hat{i} + x\hat{j}$ and the region R bounded by the unit circle C: $r(t) = (\cos t)\hat{i} + (\sin t)\hat{j}; \quad 0 \leq t \leq 2\pi$. OR Verify Green's Theorem for $\vec{F} = (x - y)i + xj$ and C is $x^2 + y^2 = 1$		7				
1020	08	Using Green's theorem evaluate $\int_C (x^2y dx + x^2 dy)$ where C is the boundary of the triangle whose vertices are (0,0), (1,0), (1,1).		7				
1021	08	Verify Green's theorem in the xy-plane for $\oint (xy + y^2)dx + x^2 dy$ where C is the closed curve of the region bounded by $y = x$ and $y = x^2$.		7				
1022	08	Verify Green's theorem for $\oint [(x^2 - 2xy)dx + (x^2y + 3)dy]$, Where, C is the boundary of the region bounded by $y = x^2$ and the line $y = x$		7				
1023	08	Using Green's theorem find $\oint_C (\frac{1}{y} dx + \frac{1}{x} dy)$ where C is the boundary of the region bounded by the parabola $y = \sqrt{x}$ and the lines $x = 1, x = 4, y = 1$		7				
1024	09	What is the degree of vertex d in graph? 	4	1	0	1	2	4
1025	09	In any graph the number of vertices of odd degree is _____.	2m	1	4m-1	2m+1	2m-1	2m
1026	09	Degree of any vertex in a simple graph with n- vertices is at the most _____.	n-1	1	n	n-1	2n-1	3n
1027	09	Maximum number of edges in any simple graph is _____	$\frac{n(n-1)}{2}$	1	$\frac{n(n-1)}{2}$	2n	$\frac{n(n-1)}{3}$	$\frac{n(n+1)}{2}$
1028	09	How many nodes are necessary to construct a graph with exactly 8 edges in which each node is of degree 2 ?	8	1	4	6	2	8
1029	09	Sum in-degrees of each vertex of G = _____ 	10	1	12	11	10	8
1030	09	A graph in which degree of each vertex is same is known as _____.	Regular graph	1	Complete graph	Directed complete graph	Wheel graph	Regular graph
1031	09	How many edges in a complete bipartite graph $K_{m,n}$?	mn	1	m	n	mn	m+n
1032	09	A complete Bipartite graph $K_{m,n}$ is regular if and only if ____	m=n	1	m=n	m>n	m<n	m=2n
1033	09	Which of the following pair are isomorphic? (1) Wheel graph with 5 vertices & K_5	Only (4)	1	Only (1)	Only (2)	Only (3)	Only (4)

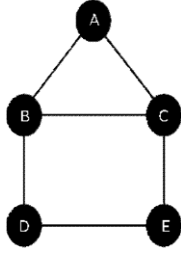
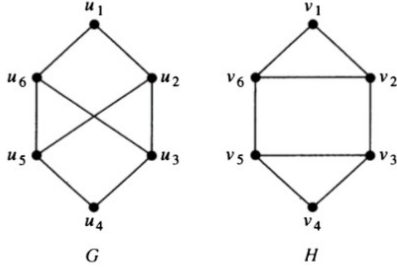
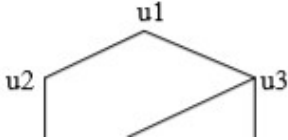
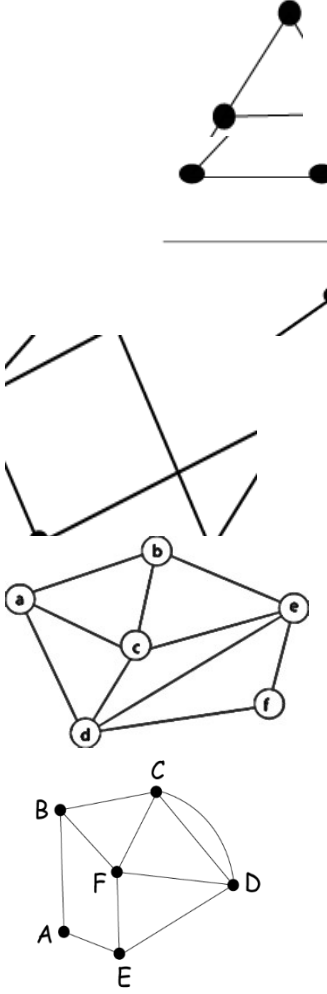
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		(2) K_6 & $K_{3,3}$ (3) C_6 & $K_{1,6}$ (4) Wheel graph with 4 vertices & K_4						
1034	09	What is the complement of graph K_{100} ?	Null graph with 100 vertices	1	$K_{1,100}$	$K_{10,10}$	C_{100}	Null graph with 100 vertices
1035	09	$C_8 \cup K_{1,8} =$ _____	wheel graph with total nine vertices	1	$K_{8,8}$	$K_{1,8}$	wheel graph with total nine vertices	C_8
1036	09	Define following graphs with specific examples (1) Null Graph (2) Loop in graph (3) Parallel Edges (4) Isolated Vertex (5) Pendant vertex		5				
1037	09	Define following types graphs with specific examples (1) Directed graph (2) Simple Graph (3) Finite Graph (4) Infinite Graph		5				
1038	09	Define following types graphs with specific examples (i) Cyclic graph (ii) Complete Graph (iii) Wheel Graph (iv) Star Graph (v) Multi Graph		5				
1039	09	For Exercises 3–to-9, determine whether the graph shown has directed or undirected edges, whether it has multiple edges, and whether it has one or more loops. Use your answers to determine the type of graph in the Table. and whether it has one or more determine the type of graph in Ta 3.  5.  7. 		7				
1040	09	Find the degree of each vertex following graphs.		5				

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		and whether it has one or more determine the type of graph in Ta 3.  5. 						
1041	09	Classify simple, multi and Pseudo graphs.  G1 G2  G3  G4  G5 		5				
1042	09	Find in degree and out degree of following graphs. G1  G2  G3 		5				
1043	09	Determine the number of vertices and edges of following graphs and, also find in degree and out degree of following graphs.		2				

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		G1G2						
1044	09	The diagram below shows a directed graph. (a) what is the in-degree of vertex c? (b) what is the out-degree of vertex d?		2				
1045	09	(A) What is the in degree of Vertex d? (B) What is the out degree of vertex c?		2				
1046	09	Draw two different graphs whose all vertices have even degree except 0 degree.		2				
1047	09	Draw following graphs and find degree of each vertex in that graph. (A) K_6 (B) $K_{1,3}$ (C) $K_{6,4}$ (D) $K_{1,1}$ (E) $K_{3,3}$ (F) $K_{5,4}$						
1048	09	Draw two different Cyclic graph, Regular graph, and Connected graphs in which both graphs contain different number of vertices. Also verify Hand Shaking lemma for them.		7				
1049	09	Verify Hand Shaking lemma for following graphs. (A)K_5 (B) C_8 (C) Wheel graph with 5 vertices (D) Star graph with 7 vertices (E) (F)		5				

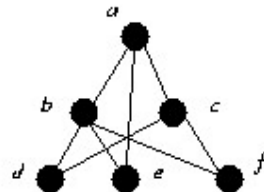
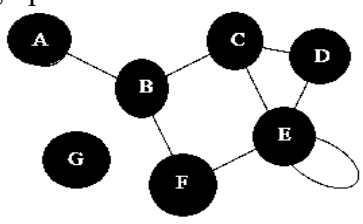
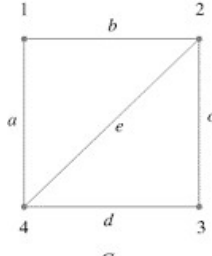
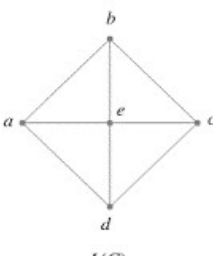
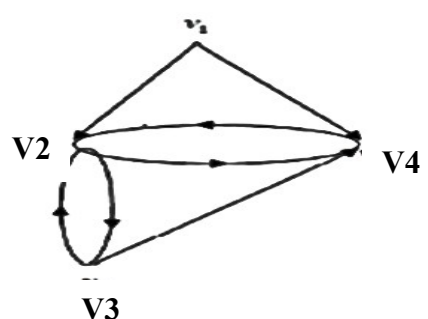
L.J Institute of Engineering and Technology, Ahmedabad. Mathematics-I Practice Book_Batch 2025 & 2026_CE Related Branches								
Note : This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.								
Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
1050	09	Verify Hand Shaking lemma for $K_{7,6}$, C_5 , Wheel graph with total no. of 7 vertices and Star graph with total no. of 7 vertices .		5				
1051	09	(1) Can a simple graph exist with 15 vertices each of degree five? (2) Show that the sum, over the set of people at a party, of the number of people a person has shaken hands with, is even. Assume that no one shakes his or her own hand.		4				
1052	09	Define Bipartite, Complete Bipartite graph, Sub graph of a graph with proper Examples.		4				
1053	09	Define path and circuit with suitable example.						
1054	09	Define isomorphic graphs. State whether the following graphs are isomorphic or not. 						
1055	09	Define isomorphic graphs. State whether the following graphs are Isomorphic or not. Justify your answer. 						
1056	09	Show that following graphs are isomorphic. 						
1057	09	Draw a graph which is regular but not bipartite.						
1058	09	Define following terms in graph with suitable examples (i) Subgraphs (ii) Spanning subgraphs (iii) Complement of a subgraph						
1059	09	Define Connected and disconnected graphs with example.						
1060	09	Find Union of graph G_1 & G_2 , Intersection of graph G_1 & G_2 and Ring sum of G_1 & G_2 .						

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
								
1061	09	Prove that (i) $K_{1,5} \cup C_5$ = Wheel graph with total 6 vertices (ii) $K_{1,4} \cup C_4$ = Wheel graph with total 5 vertices		3				
1062	09	Find the three proper sub graphs of graph G with minimum two vertices and complement of these subgraphs. 		5				
1063	09	Find the path and circuit, in following graph. 						
1064	09	Write three different path and circuit in following graph, if exist.		7				

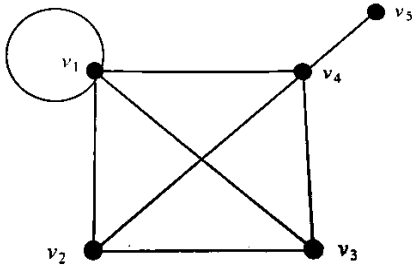
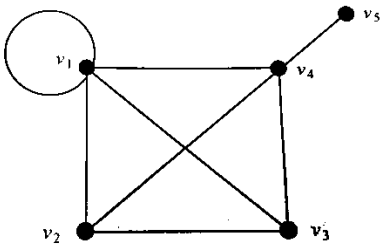
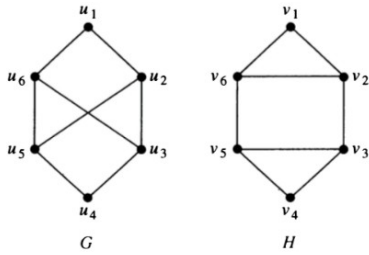
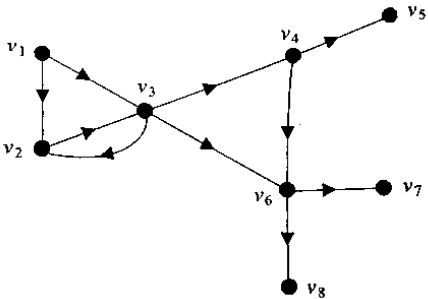
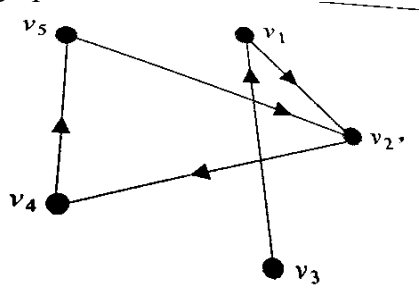
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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
		 						
1065	09	Define Unicursal line (Open Walk with every edges) and Euler Line (Closed Walk with every edges). And find it if possible. If Unicursal line or Euler line is not existed in a graph then give its proper reason. 		5				
1066	09	Define Hamilton path & circuit with proper example.		4				
1067	09	Find Hamilton path & circuit in the graph if possible. 		10				
1068	09	From the given graph, the Eccentricity of Vertex “e” is.....	4	1	2	3	5	4

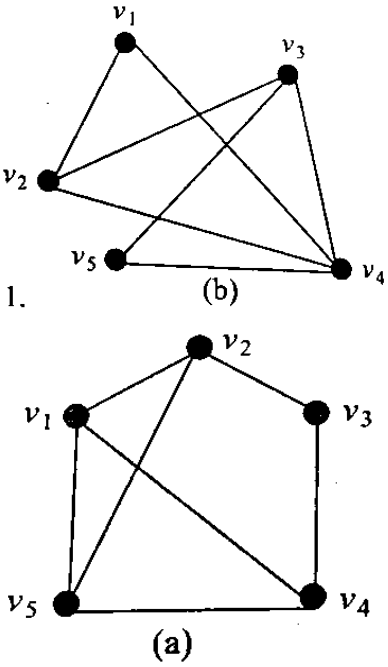
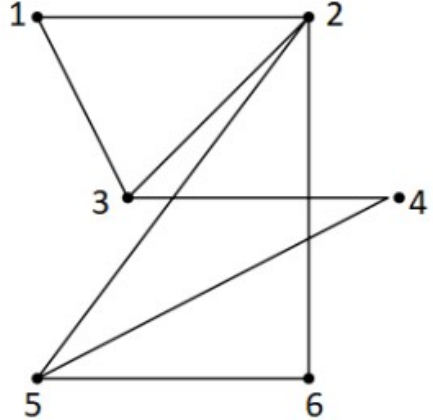
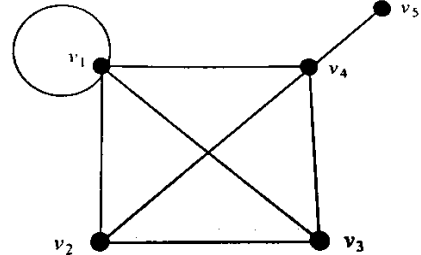
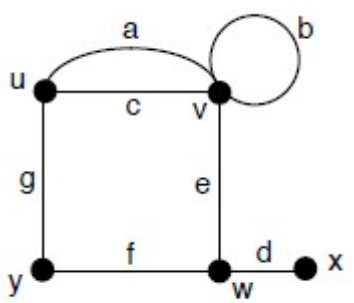
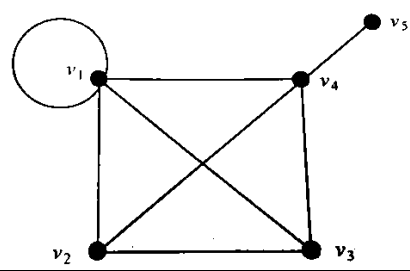
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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
1069	09	From the given graph, the radius is.....	3	1	2	3	4	5
1070	09	From the given graph, the diameter is.....	6	1	3	6	4	5
1071	09	Which of the following graphs have Unicursal line or Euler line?		4				
1072	09	Show any one Hamiltonian circuit.		4				
1073	09	Using Dijkstra's Algorithm find the shortest path between 0 to 7.		7				
1074	09	Using Dijkstra's Algorithm find the shortest path between (i)A to F (ii) A to E		7				
1075	09	Find the smallest path for the Salesman such that he covered each city exactly once and return to staring city.		7				

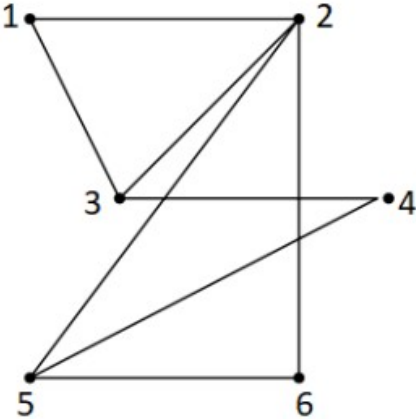
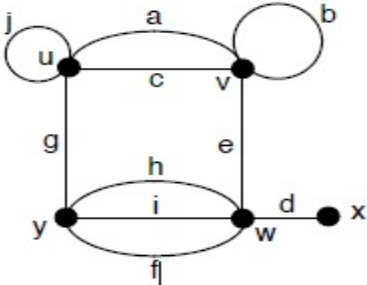
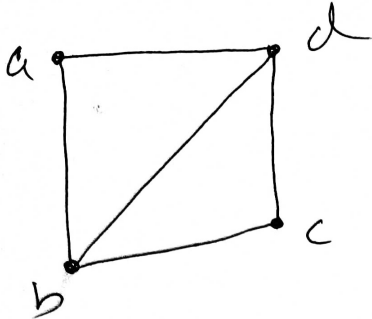
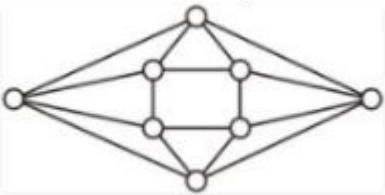
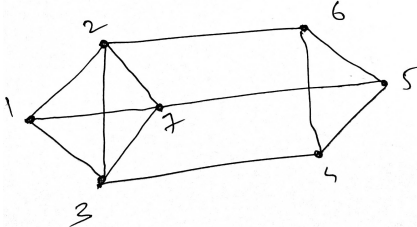
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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
1076	09	Find the smallest path for the Salesman such that he covered each city exactly once and return to starting city A.		7				
1077	09	Define Cut vertex and Cut sets with suitable examples.		3				
1078	09	Find $(G-\{v\})$, $(G-\{e\})$ & $G-\{f, g\}$ for the graph G.		3				
1079	09	Find all cut vertices and cut sets in following graph.		4				
1080	09	Find the cut vertices and cut sets for the following graphs		4				
1081	09	Find the cut vertices and cut sets for the following graphs		4				

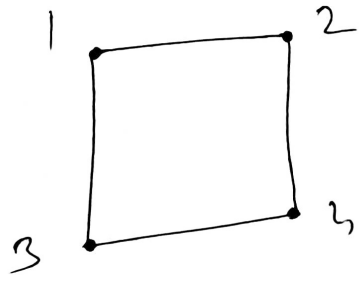
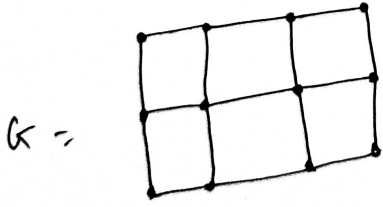
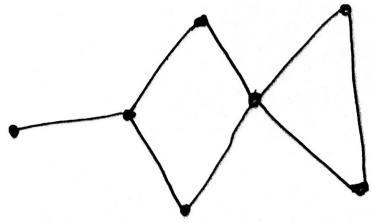
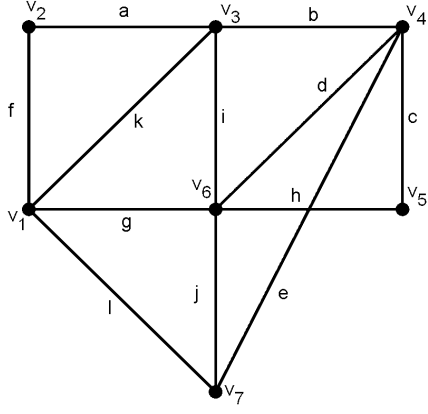
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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D																																				
1082	09	Find the following from given graph, cut set of size 2, or explain why no such exists.  G_2		4																																								
1083	09	Define Adjacency matrix and path matrix of a graph. Explain each with example.		3																																								
1084	09	Find the adjacency matrix for the following graphs.   G  $L(G)$																																										
1085	09	From the matrix $X(G) = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$ draw a graph G and find the degree of each vertex of graph.		3																																								
1086	09	From the adjacency matrix draw a directed graph and find in and out degree of each vertex. <table><tr><td></td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td></tr><tr><td>A</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>B</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>C</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>D</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>E</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>		A	B	C	D	E	A	0	1	1	0	0	B	0	0	0	1	1	C	0	0	0	1	0	D	1	0	0	1	1	E	0	0	0	0	0		3				
	A	B	C	D	E																																							
A	0	1	1	0	0																																							
B	0	0	0	1	1																																							
C	0	0	0	1	0																																							
D	1	0	0	1	1																																							
E	0	0	0	0	0																																							
1087	09	From the adjacency matrix $X(G) = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix}$ draw a graph G and find the degree of each vertex of graph.		4																																								
1088	09	Define adjacency matrix of a digraph. Obtain the adjacency matrix A of the given digraph. 		7																																								

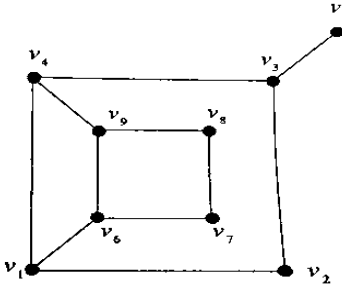
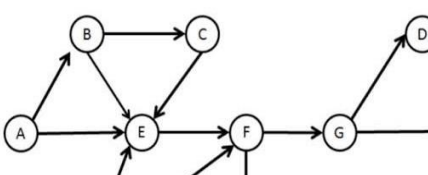
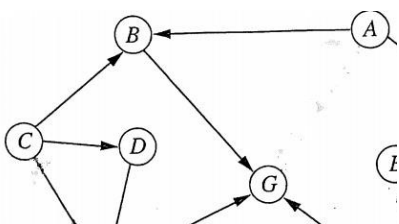
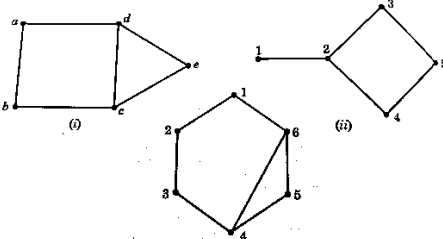
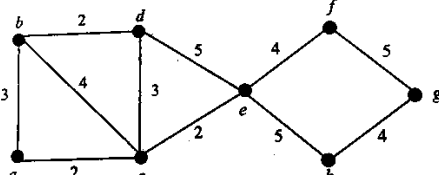
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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
1089	09	Write the Adjacency matrix in following graphs. (A) K_4 (B) $K_{1,3}$ (C) $K_{2,3}$ (D) $K_{3,3}$		7				
1090	09	Determine the matrices of adjacency and incidence for the given graph G.		3				
1091	09	Determine the matrices of adjacency and incidence for the given graph G.		3				
1092	09	Determine the matrices of adjacency and incidence for the given graph G.		3				
1093	09	Determine the matrices of adjacency for the given graph G.		3				
1094	09	Draw a graph that has the given adjacency matrix. $X(G) = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$		3				
1095	09	Find the Adjacency matrix for the given graph:		2				

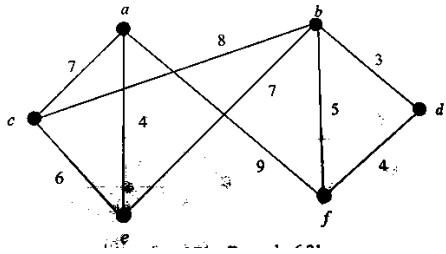
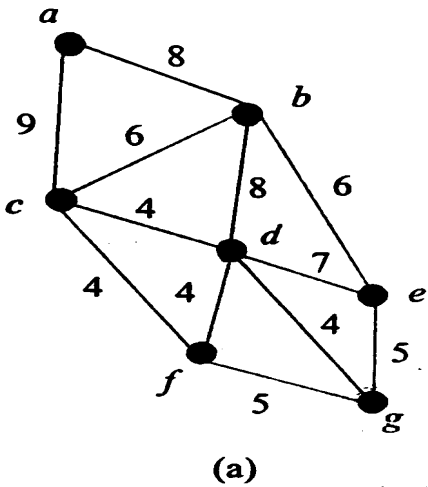
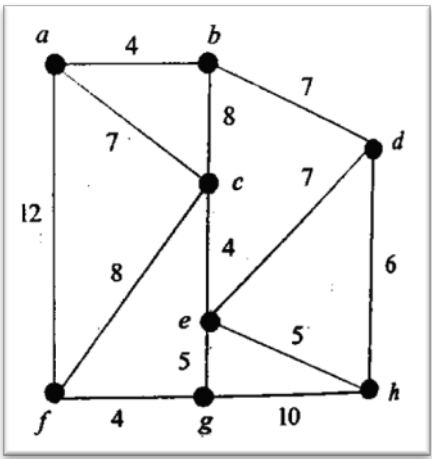
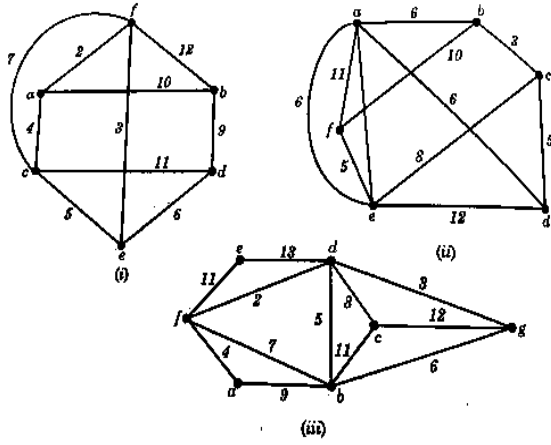
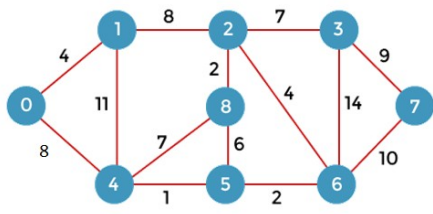
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1096	09	Draw the Digraph with the given matrix as adjacency matrix: $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$		2				
1097	09	Draw the Digraph with the given matrix as adjacency matrix: $\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$		2				
1098	09	Find the Adjacency matrix for the given graph: 						
1099	09	Find the circuit matrix for graph and path matrix between v_2 and v_5 . 						
1100	09	Find the circuit matrix for graph G and H and path matrix between i) u_1 and u_4 ii) v_6 and v_3 . 						
1101	09	Find the Adjacency matrix for the given graph: 						
1102	09	Find the Adjacency matrix for the given graph. 						
1103	09	Find the adjacency matrix for the following graphs:		4				

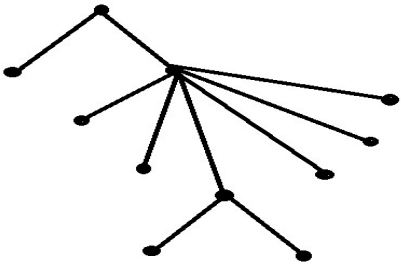
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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
								
1104	09	Find adjacency matrix, incidence matrix and path matrix between vertices (1,5). 						
1105	09	Find the cut set matrix for the following graph. 						
1106	09	Find the cut set matrix for the following graph. 						
1107	09	Find the Fundamental circuit matrix for following graph 						
1108	09	Find the Fundamental circuit matrix for following graph						

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1109	09	Find the Fundamental circuit matrix for following graph 						
1110	09	If $E = \emptyset$ then $\chi(G) =$ _____	1	1	1	0	-1	-2
1111	09	What is the chromatic number of simple graph G, if $E \neq \emptyset$?	$\chi(G) \geq 2$	1	$\chi(G) \geq 2$	$\chi(G) \geq 1$	$\chi(G) \geq 3$	$\chi(G) \geq 4$
1112	09	$\chi(G) =$ _____; if $G = K_n$	n	1	-n	0	n	2n
1113	09	$\chi(G) =$ _____; if $G = C_n$ and n is odd.	3	1	0	1	2	3
1114	09	$\chi(G) =$ _____; if $G = C_n$ and n is even.	2	1	0	1	2	3
1115	09	If G is a tree then $\chi(G) =$ _____;	2	1	0	1	2	3
1116	09	Every graph having Triangle is at least _____	3- chromatic	1	0- chromatic	1- chromatic	2- chromatic	3- chromatic
1117	09	Define chromatic number? Find it for $K_{7,5}$.		3				
1118	09	Find the Chromatic number of the Graph 		3				
1119	09	Find the minimum number of colors required to color the following graph, such that no two adjacent vertices are assigned the same color. 		4				
1120	09	Find the Chromatic number of the Graph 		3				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
1121	09	Find the Chromatic polynomial of the Graph 		3				
1122	09	What is the chromatic number of peterson Graph?		3				
1123	09	$\chi(G) = \underline{\hspace{2cm}}$; if G is the following graph. 		4				
1124	09	$\chi(G) = \underline{\hspace{2cm}}$; if G is the following graph. 		3				
1125	09	Find cut sets with 4 edges. 		5				
1126	10	Let $G = (V, E)$ be any connected undirected edge-weighted graph. The weights of the edges in E are positive. Consider the following statements: The path between a pair of vertices in a minimum spanning tree of an undirected graph is necessarily the shortest (minimum weight) path. Minimum Spanning Tree of G is always unique and shortest path between a pair of vertices may not be unique. Which of the above statements is/are necessarily true?	Neither (1) nor (2)	1	Only (1)	Only (2)	Both (1) and (2)	Neither (1) nor (2)
1127	10	Undirected graph G has n nodes. Its adjacency matrix is given by $n \times n$ square matrix whose (1) Diagonal elements are 0's. (2) Non-diagonal elements are 1's. Which of the following one is true?	Graph G has multiple distinct MST's each of cost $n - 1$	1	Graph G has no minimum spanning tree (MST)	Graph G has unique MST's each of cost $n - 1$	Graph G has multiple distinct MST's each of cost $n - 1$	Graph G has multiple spanning tree of different costs
1128	10	What would be the number of zeros in the adjacency matrix of the given graph? 	10	1	10	6	16	0

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1129	10	List all the paths from v_2 to v_9 for the given Graph (a).  (a)		2				
1130	10	Define with example Centre of a graph and radius of a tree.		2				
1131	10	A tree T has 4 vertices of degree 2, 4 vertices of degree 3, 2 vertices of degree 4. Find the number of pendent vertices in T.		3				
1132	10	 For the given Diagram find the following answers: (a) How many paths exist from the Vertex A and vertex K? Justify your answer by listing the paths (b) Is the graph weakly Connected? Justify your answer.		4				
1133	10	Find the all the possible paths from the vertex A to E, B to G. Is it Unilaterally connected graph or Weakly connected or both? 		4				
1134	10	Show that it is a tree with 10 vertices which has vertices either of degree 1 or of degree 3 and draw the tree. Is it possible to draw the same type of tree with 11 vertices?		4				
1135	10	Construct three different spanning tree for the following graphs. 		4				
1136	10	Find the Minimal Spanning tree for the following weighted connected graph: 		5				
1137	10	Using Prim's Algorithm, find the minimum weight spanning tree for the following weighted connected graph:		5				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
								
1138	10	Find the Minimal Spanning tree for the following weighted connected graph using Prim's Algorithm:  (a)		5				
1139	10	Using Prim's Algorithm, find the minimum weight spanning tree for the following weighted connected graph: 		5				
1140	10	Use Prim's Algorithm to determine the minimal spanning tree for the following graphs. 		5				
1141	10	Using Prim's Algorithm, find the minimum weight spanning tree for the following weighted connected graph (Initial vertex is "0") 		5				

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1142	10	 The path length of above tree is	None of these	1	19	20	21	None of these
1143	10	Total no. of Spanning tree in K_n is _____	n^{n-2}	1				
1144	10	A graph is Connected iff it has a _____	At least one path between each pair of vertices	1				
1145	10	Total no. of Spanning tree in K_6 is _____	6^4	1				
1146	10	Draw all the spanning tree of K_3		2				
1147	10	Find two possible spanning trees from below graph.		2				
1148	10	Which of the following is/are true in the case of a spanning tree of a graph G? A) It is a subgraph of the G. B) It includes every vertex of the G. C) It is cyclic.		1				